

Data Analysis

Import Packages

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Read the File

```
In [2]: file_path=r"C:\Users\Lenovo\Desktop\Gen ai\categorical pro\categorical pro\VisaD
visa_df = pd.read_csv(file_path)
visa_df
```

```
Out[2]:
```

	case_id	continent	education_of_employee	has_job_experience	requires_job_t
0	EZYV01	Asia	High School		N
1	EZYV02	Asia	Master's		Y
2	EZYV03	Asia	Bachelor's		N
3	EZYV04	Asia	Bachelor's		N
4	EZYV05	Africa	Master's		Y
...	...	...	...	...	...
25475	EZYV25476	Asia	Bachelor's		Y
25476	EZYV25477	Asia	High School		Y
25477	EZYV25478	Asia	Master's		Y
25478	EZYV25479	Asia	Master's		Y
25479	EZYV25480	Asia	Bachelor's		Y

25480 rows × 12 columns



Frequency Table

- can be done on categorical value

```
In [3]: cat = visa_df.select_dtypes(include=['object', 'category']).columns
num=visa_df.select_dtypes(include=['int64', 'float64']).columns
```

```
cat
num
```

```
Out[3]: Index(['no_of_employees', 'yr_of_estab', 'prevailing_wage'], dtype='object')
```

Continental

```
In [4]: visa_df['continent'] # selecting one column
```

```
Out[4]: 0      Asia
1      Asia
2      Asia
3      Asia
4      Africa
...
25475   Asia
25476   Asia
25477   Asia
25478   Asia
25479   Asia
Name: continent, Length: 25480, dtype: object
```

```
In [5]: visa_df['continent'].values
```

```
Out[5]: array(['Asia', 'Asia', 'Asia', ..., 'Asia', 'Asia', 'Asia'], dtype=object)
```

```
In [6]: visa_df[['continent']] # it is data frame ----> bcz[[]]
```

```
Out[6]:
```

	continent
0	Asia
1	Asia
2	Asia
3	Asia
4	Africa
...	...
25475	Asia
25476	Asia
25477	Asia
25478	Asia
25479	Asia

25480 rows × 1 columns

unique

- how many different continent are there uses unique

visa_df -----> Total data frames or data set

visa_df ['continent'] -----> selecting one column

visa_df['continent'] . unique() -----> selecting unique columns

visa_df['continent'].nunique() -----> n-unique()

nunique

```
In [7]: visa_df['continent'].unique()
```

```
Out[7]: array(['Asia', 'Africa', 'North America', 'Europe', 'South America',  
              'Oceania'], dtype=object)
```

```
In [8]: visa_df['continent'].nunique() ## how many continents are unique is 6
```

```
Out[8]: 6
```

Frequency Table

Total no-of classes=6

Total no-of rows:25480

Different count of visa application are from different companies

How many applications are from asia

```
In [9]: visa_df['continent']=="Asia"
```

```
Out[9]: 0      True
        1      True
        2      True
        3      True
        4     False
        ...
        25475   True
        25476   True
        25477   True
        25478   True
        25479   True
        Name: continent, Length: 25480, dtype: bool
```

```
In [10]: con=visa_df['continent']=="Asia"
        con
```

```
Out[10]: 0      True
         1      True
         2      True
         3      True
         4     False
         ...
         25475   True
         25476   True
         25477   True
         25478   True
         25479   True
         Name: continent, Length: 25480, dtype: bool
```

```
In [11]: visa_df[con]
```

```
Out[11]:
```

	case_id	continent	education_of_employee	has_job_experience	requires_job_1
0	EZYV01	Asia	High School		N
1	EZYV02	Asia	Master's		Y
2	EZYV03	Asia	Bachelor's		N
3	EZYV04	Asia	Bachelor's		N
5	EZYV06	Asia	Master's		Y
...	...	...	...	...	...
25475	EZYV25476	Asia	Bachelor's		Y
25476	EZYV25477	Asia	High School		Y
25477	EZYV25478	Asia	Master's		Y
25478	EZYV25479	Asia	Master's		Y
25479	EZYV25480	Asia	Bachelor's		Y

16861 rows × 12 columns



```
In [12]: len(visa_df[con])
```

Out[12]: 16861

- Steps:
- 1.visa_df
- 2.visa_df['continent']=="Asia"
- 3.con=visa_df['continent']=="Asia"
- 4.visa_df[con]
- 5.len(visa_df[con])

```
In [13]: con=visa_df['continent']=="Asia"
len(visa_df[con])

con=visa_df['continent']=="Africa"
len(visa_df[con])

con=visa_df['continent']=="Asia"
len(visa_df[con])
```

Out[13]: 16861

- As the python repeats the step 12 times
- i get from unique
- So taking 'i'

Task

unique

empty list=[]

for loop

append the length

```
In [14]: labels=visa_df['continent'].unique()
count = []
for i in labels:

    cou=visa_df['continent']==i
```

```

values=len(visa_df[con])
count.append(values)

print(labels)
print(count)

```

```

['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
[16861, 16861, 16861, 16861, 16861, 16861]

```

```

In [15]: ft=pd.DataFrame(zip(labels,count),
                        columns=['class','Frequency'])
ft.to_csv('continent_frequency.csv',index=False)

```

Value Counts

```

In [16]: # short cut
visa_df['continent'].value_counts()

```

```

Out[16]: continent
Asia                16861
Europe              3732
North America       3292
South America        852
Africa               551
Oceania              192
Name: count, dtype: int64

```

```

In [17]: labels=visa_df['continent'].value_counts().keys()
count=visa_df['continent'].value_counts().values

```

```

In [18]: visa_df['continent'].value_counts()
labels=visa_df['continent'].value_counts().keys()
count=visa_df['continent'].value_counts().values
labels

```

```

Out[18]: Index(['Asia', 'Europe', 'North America', 'South America', 'Africa',
               'Oceania'],
              dtype='object', name='continent')

```

```

In [19]: count

```

```

Out[19]: array([16861, 3732, 3292, 852, 551, 192])

```

BAR CHART

```

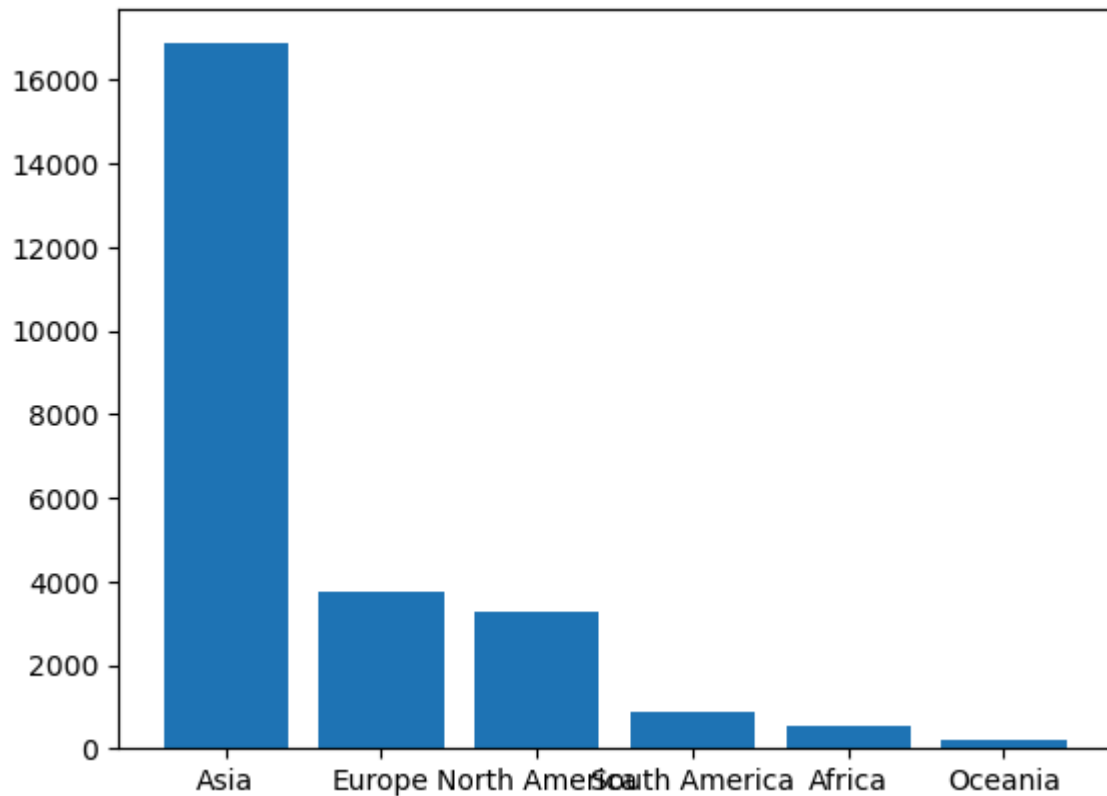
In [20]: plt.bar(labels,count)

```

```

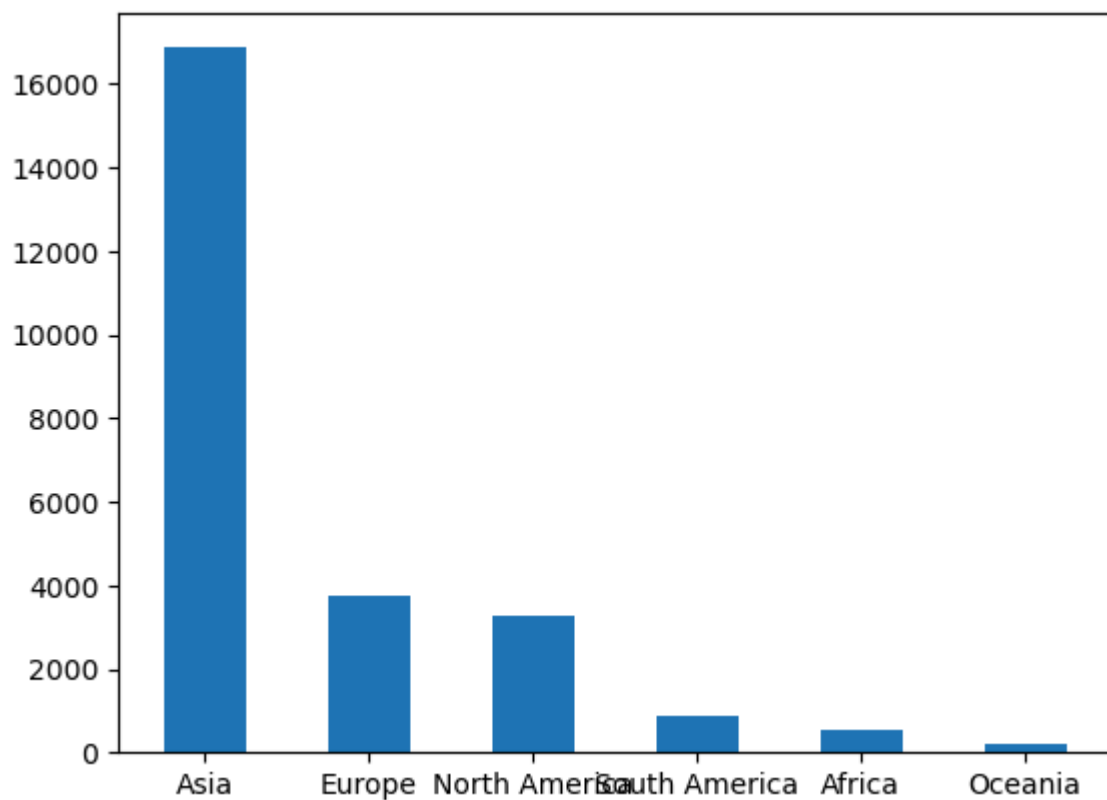
Out[20]: <BarContainer object of 6 artists>

```



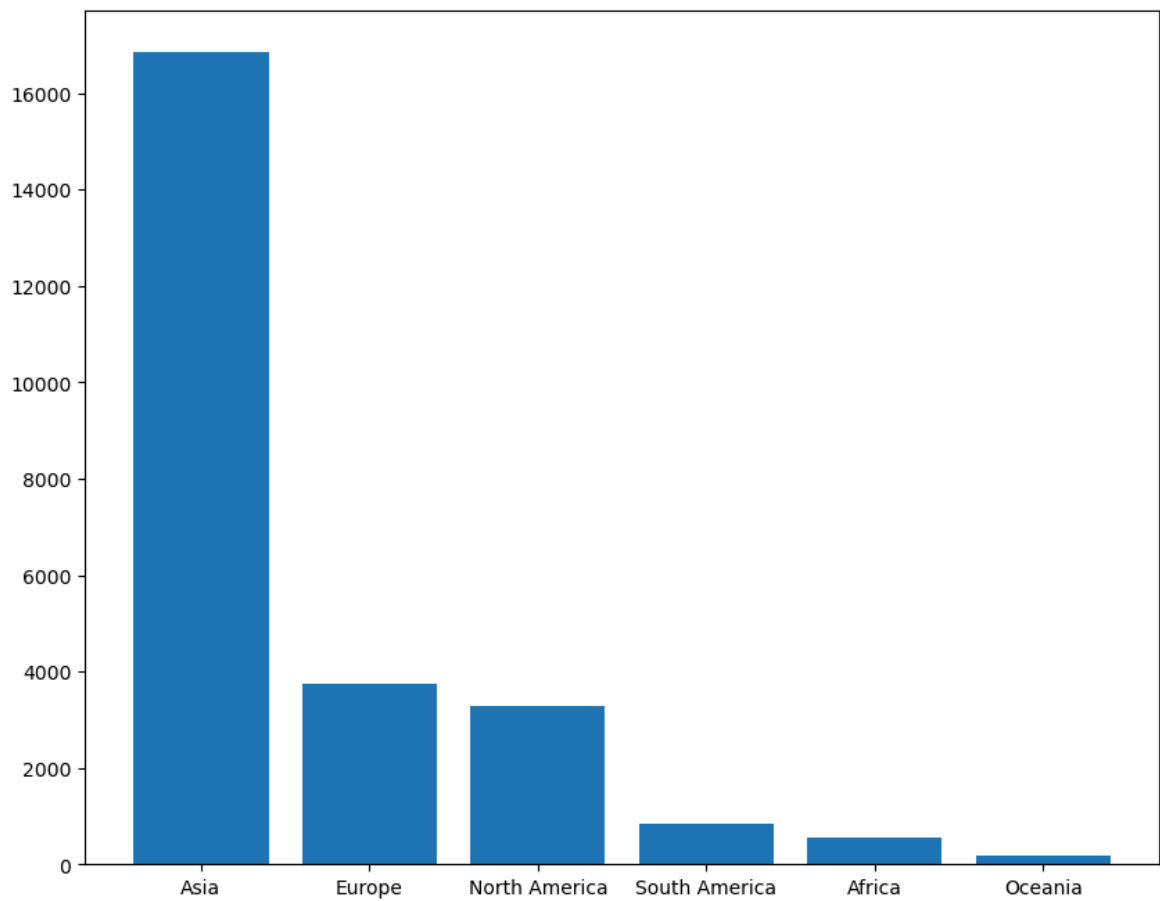
```
In [21]: plt.bar(labels,count,width=0.5) ## width -----> bar width
```

```
Out[21]: <BarContainer object of 6 artists>
```



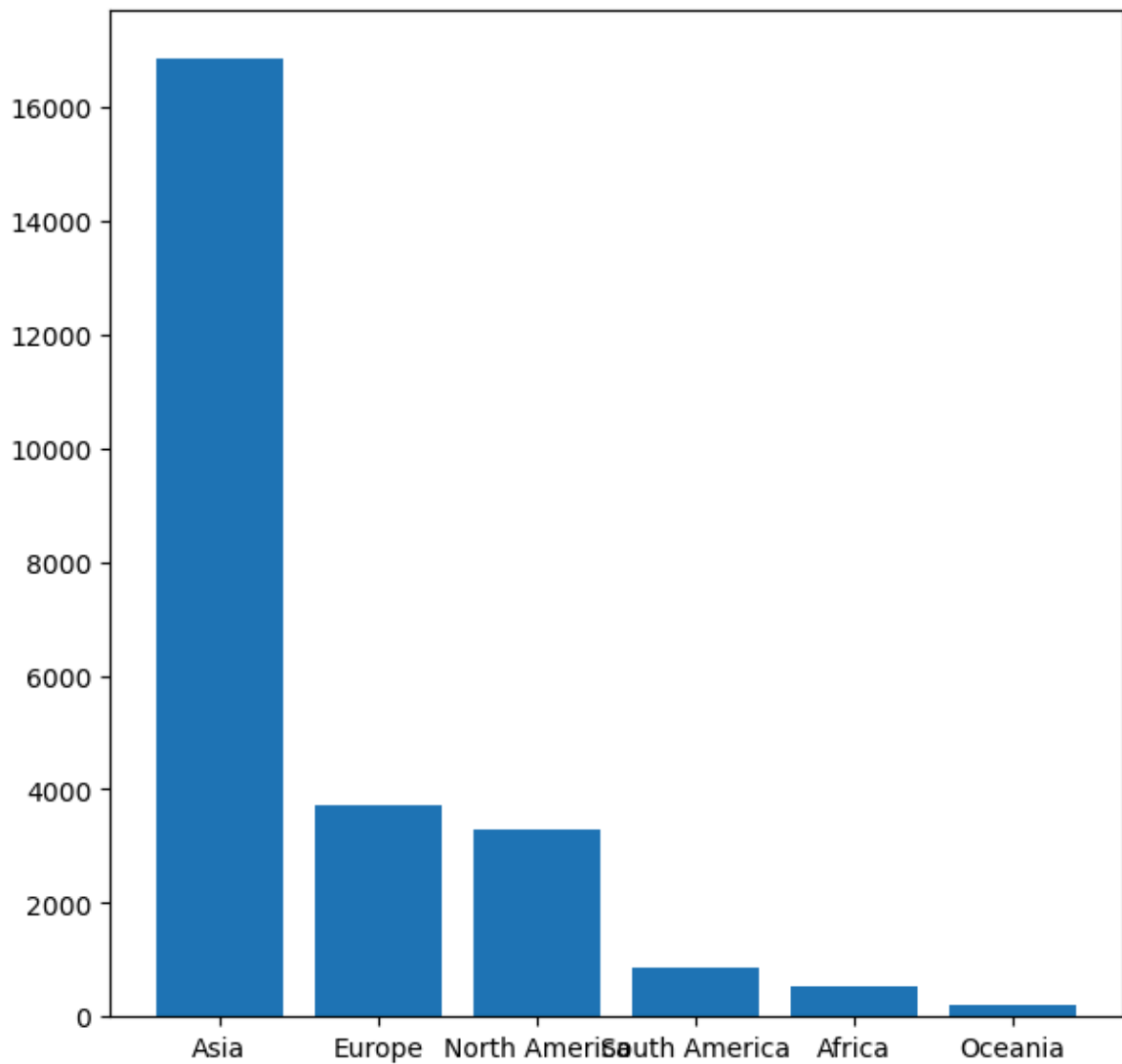
```
In [22]: plt.figure(figsize=(10,8))  
plt.bar(labels,count)
```

```
Out[22]: <BarContainer object of 6 artists>
```



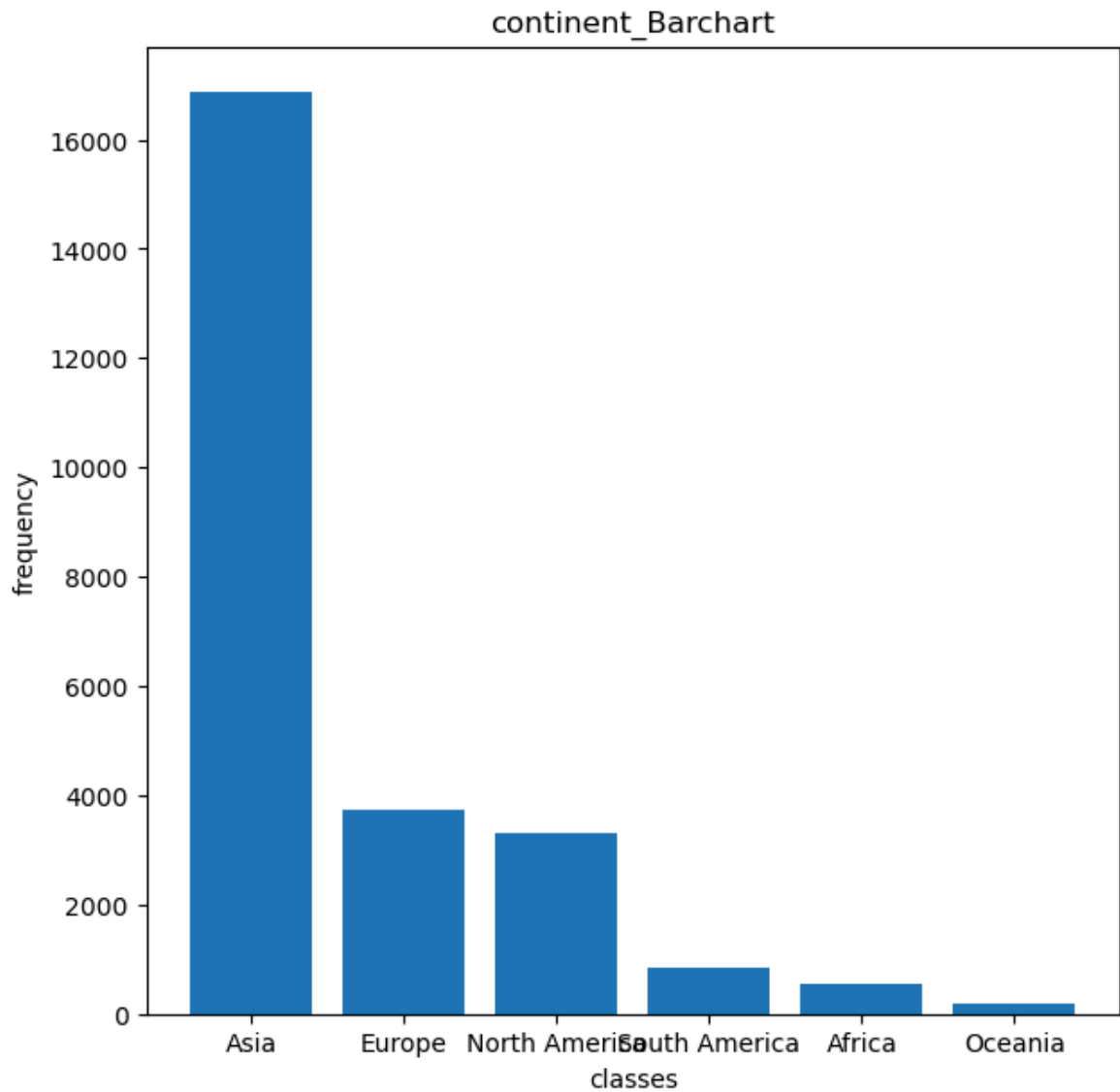
```
In [23]: plt.figure(figsize=(7,7)) ## 10--x--axis,,, 8-y-axis  
plt.bar(labels,count)
```

```
Out[23]: <BarContainer object of 6 artists>
```

```
In [24]: plt.figure(figsize=(7,7)) ## 10--x--axis,,, 8-y-axis
plt.bar(labels,count)
plt.xlabel('classes')
plt.ylabel('frequency')
plt.title("continent_Barchart")
```

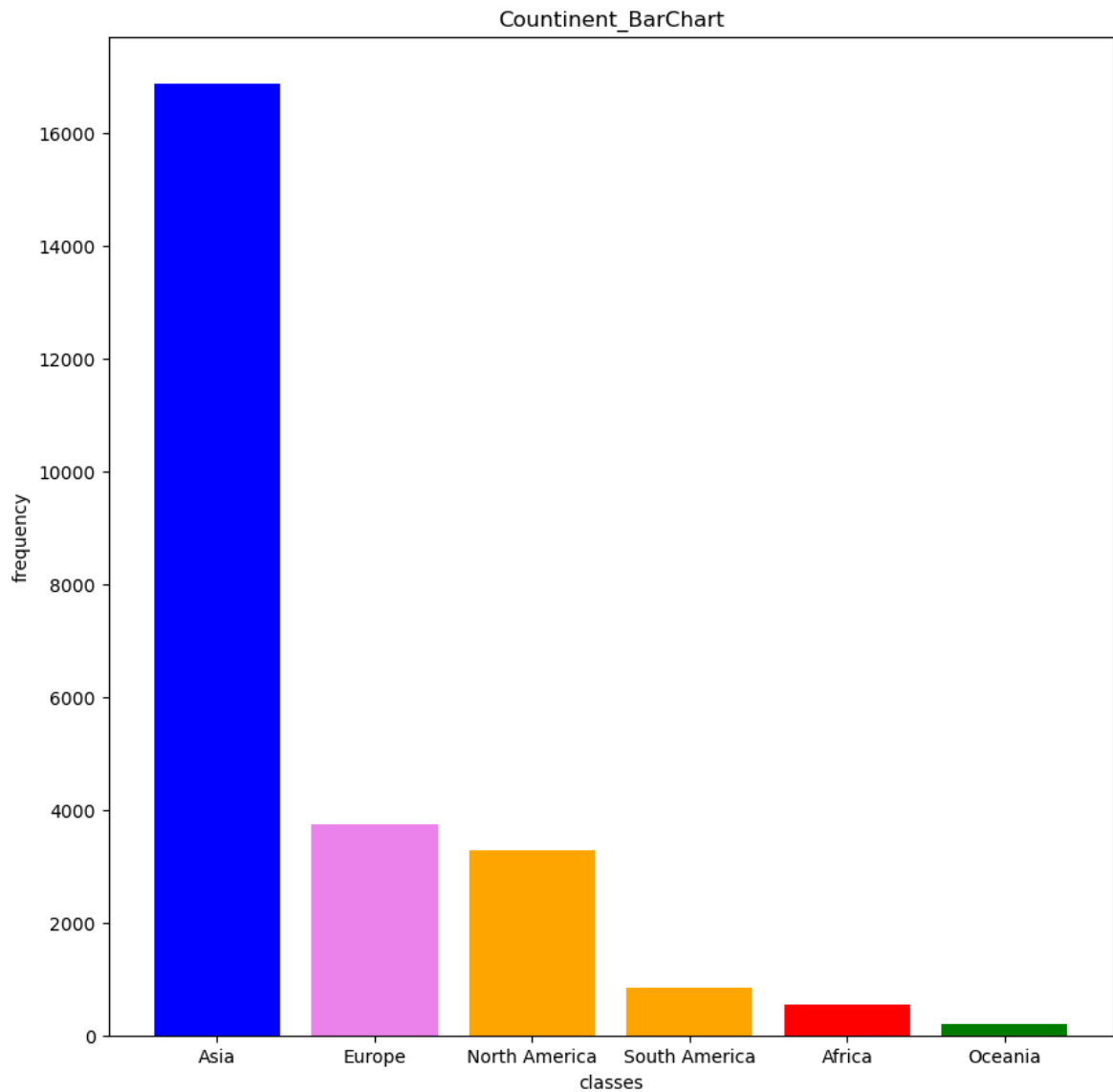
```
Out[24]: Text(0.5, 1.0, 'continent_Barchart')
```



```
In [25]: plt.figure(figsize=(10,10))

col=['blue','violet','orange','orange','red','green']
plt.bar(labels,count,color=col)
plt.bar(labels,count,color=col)
plt.xlabel('classes')
plt.ylabel('frequency')
plt.title('Countinent_BarChart')
```

```
Out[25]: Text(0.5, 1.0, 'Countinent_BarChart')
```



```
In [26]: # saving the graph
plt.savefig('continent_barchart.jpg')
```

<Figure size 640x480 with 0 Axes>

```
In [27]: plt.show()

# no lines will be executed
# saving is done before plt.show()
```

PIE CHART -----> uses percentages

RELATIVE FREQUENCY TABLE

```
In [28]: visa_df['continent'].value_counts(normalize=True)/100
```

```
Out[28]: continent
Asia      0.006617
Europe    0.001465
North America 0.001292
South America 0.000334
Africa     0.000216
Oceania    0.000075
Name: proportion, dtype: float64
```

```
In [29]: visa_df['continent'].value_counts(normalize=True)*100
```

```
Out[29]: continent
Asia      66.173469
Europe    14.646782
North America 12.919937
South America 3.343799
Africa     2.162480
Oceania    0.753532
Name: proportion, dtype: float64
```

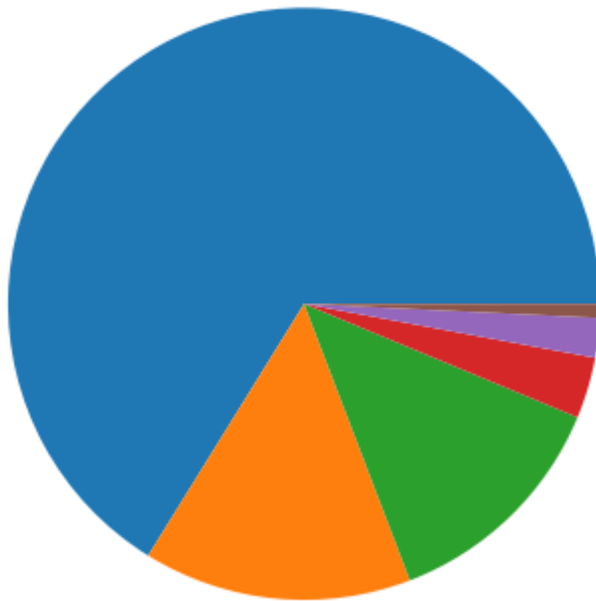
PIE CHART

pie chart also require keys and values

```
In [30]: keys=visa_df['continent'].value_counts().keys()
values=visa_df['continent'].value_counts().values
```

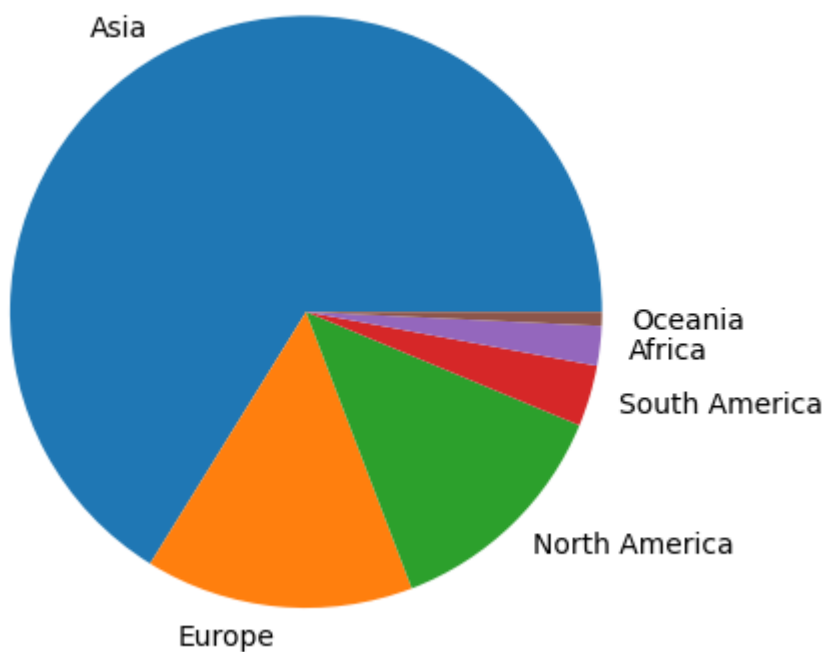
```
In [31]: plt.pie(values)
```

```
Out[31]: ([<matplotlib.patches.Wedge at 0x1c8bba89010>,
<matplotlib.patches.Wedge at 0x1c8bca40cd0>,
<matplotlib.patches.Wedge at 0x1c8bca40f50>,
<matplotlib.patches.Wedge at 0x1c8bca411d0>,
<matplotlib.patches.Wedge at 0x1c8bca41450>,
<matplotlib.patches.Wedge at 0x1c8bca416d0>],
[Text(-0.5351742797933029, 0.9610351139514726, ''),
Text(-0.10373483734065869, -1.0950977506697324, ''),
Text(0.7670028608214846, -0.7884837420591869, ''),
Text(1.054611802949007, -0.31272023452383935, ''),
Text(1.092698639567901, -0.12652937637741896, ''),
Text(1.099691791437175, -0.02603773887449993, '')[0]])
```



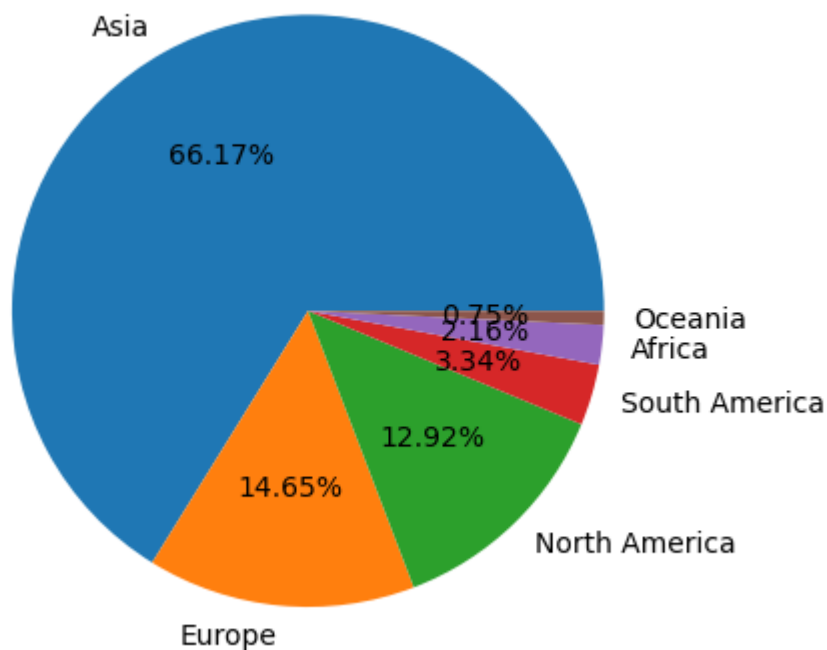
```
In [32]: plt.pie(values, labels=keys)
plt.show()

# plt.show() clear the data which we got above
```

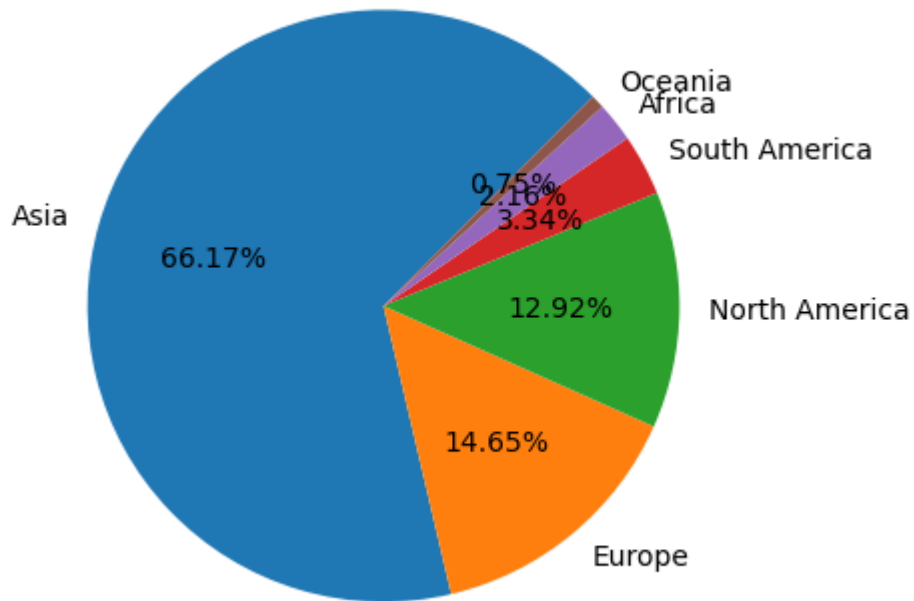


- autopct is a percentage
- autopct == %1.0f%%----->25%
 - 1.1f%----->25.1%(1----->after takes one digit)
 - 1.2f%----->25.34%(2----->after takes 2 digits after decimal)
 - 1.3f%----->25.456%(3--->3 digits)

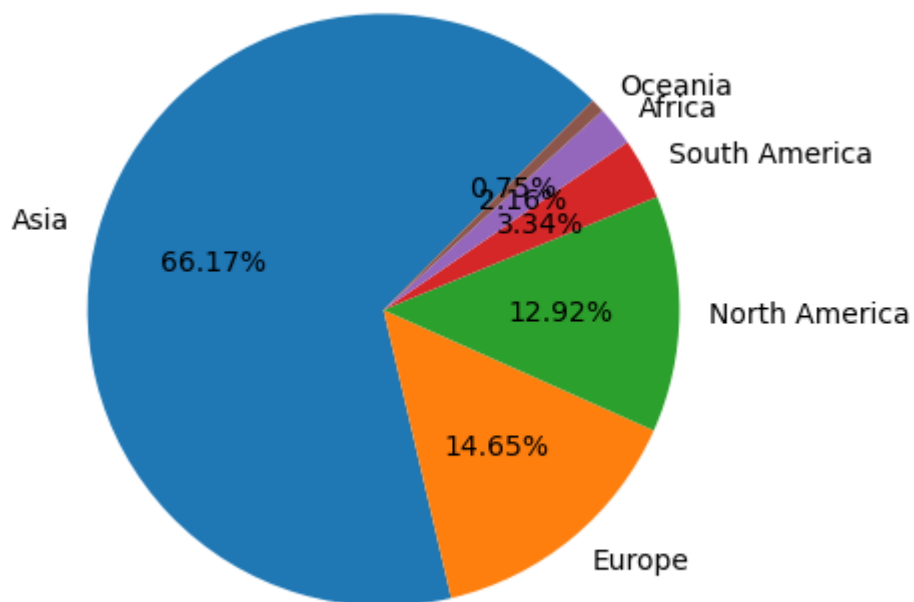
```
In [33]: plt.pie(values, labels=keys, autopct='%1.2f%%') # autopct =====is string
plt.show()
```



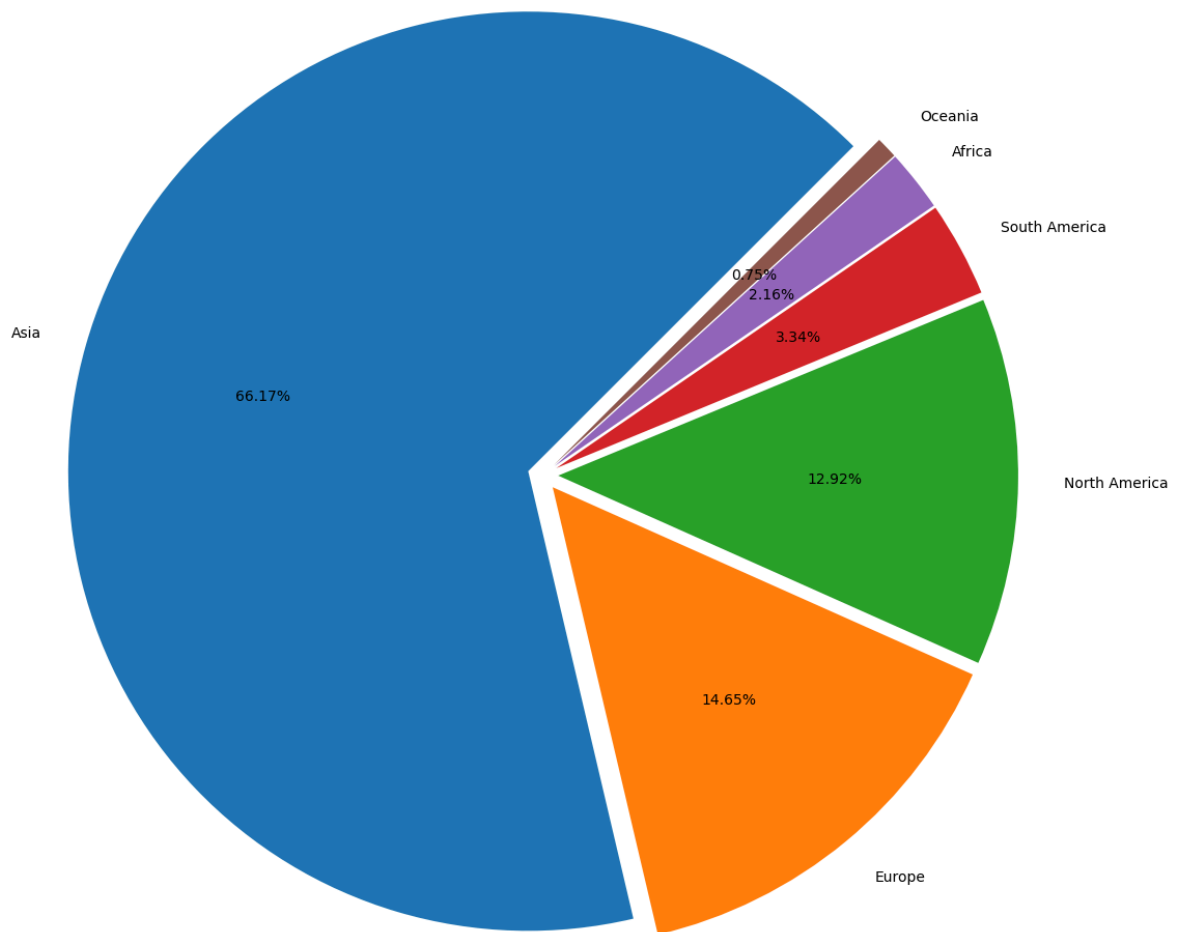
```
In [34]: plt.pie(values, labels=keys, autopct='%1.2f%%', startangle=45) # autopct====is stri
plt.show()
```



```
In [35]: plt.pie(values, labels=keys, startangle=45, autopct='%1.2f%%') #atopct == is stri
plt.show()
```



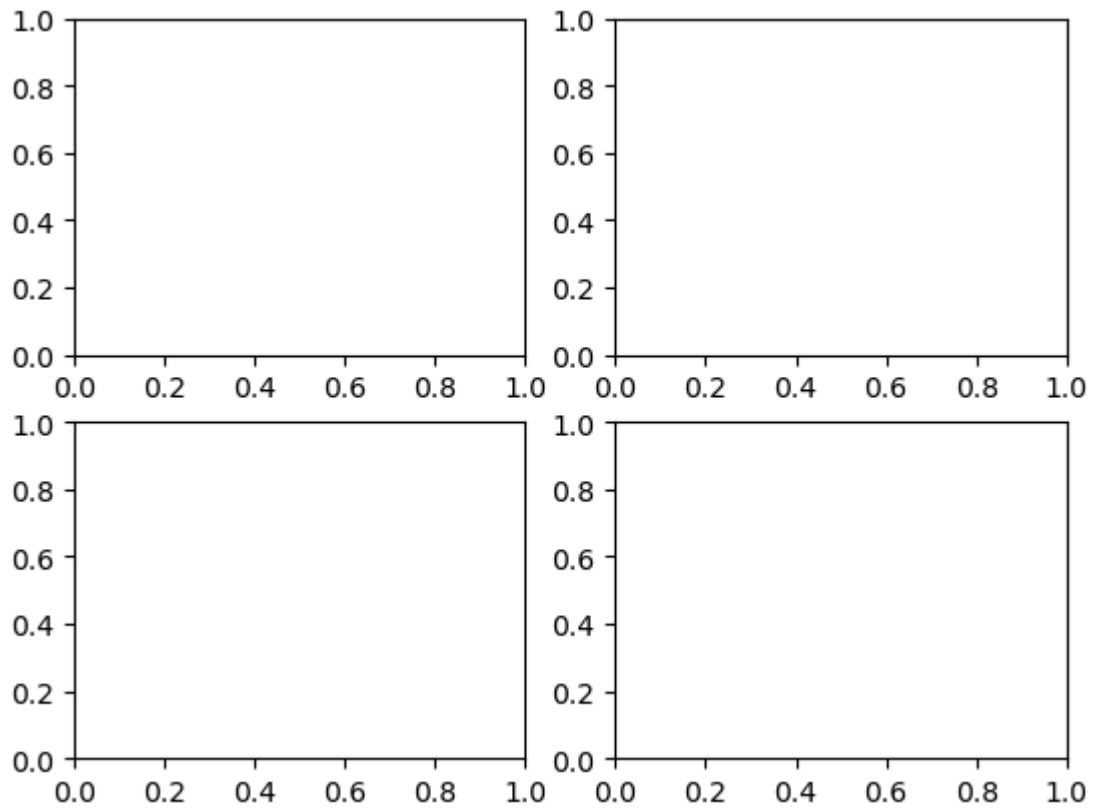
```
In [36]: plt.pie(values, labels=keys, startangle=45, autopct='%1.2f%%', explode=[0.1,0.1,0.1],
plt.show()
```



SUB - PLOTS

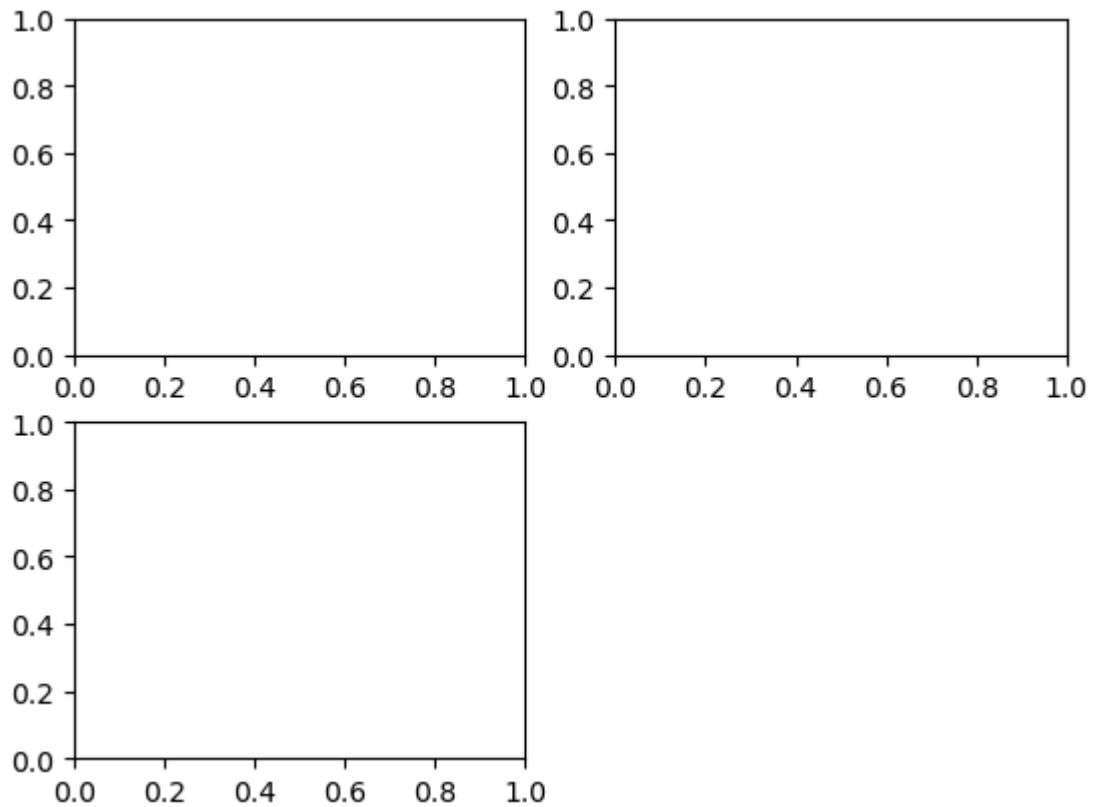
```
In [37]: plt.subplot(2,2,1)
plt.subplot(2,2,2)
plt.subplot(2,2,3)
plt.subplot(2,2,4)
```

Out[37]: <Axes: >



```
In [38]: plt.subplot(2,2,1)
plt.subplot(2,2,2)
plt.subplot(2,2,3)
```

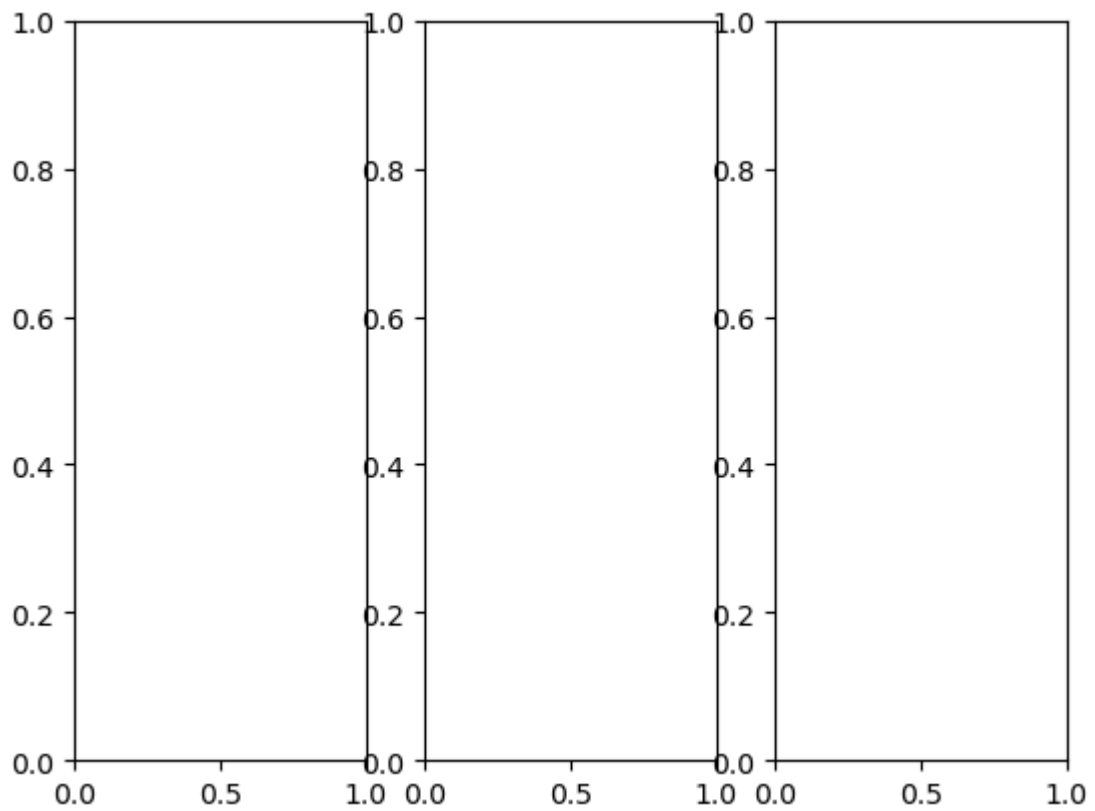
Out[38]: <Axes: >



```
In [39]: plt.subplot(1,3,1)
plt.subplot(1,3,2)
```

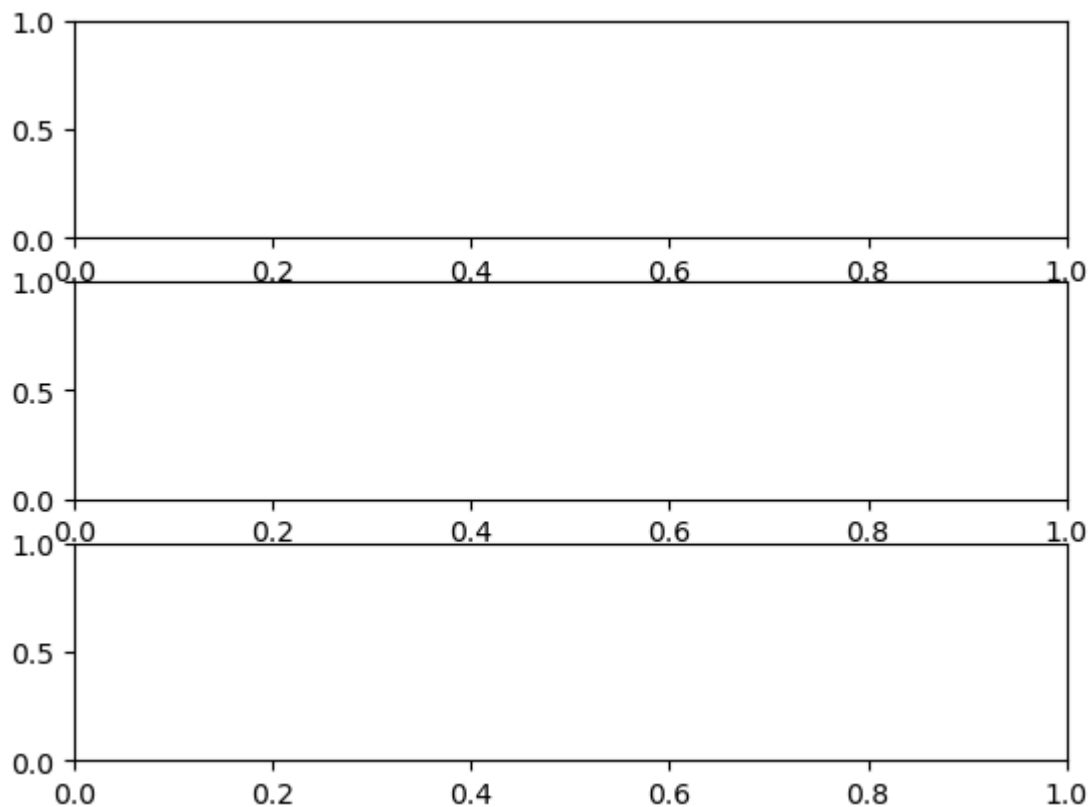
```
plt.subplot(1,3,3)  
#<3 only
```

Out[39]: <Axes: >



```
In [40]: plt.subplot(3,1,1)  
plt.subplot(3,1,2)  
plt.subplot(3,1,3)
```

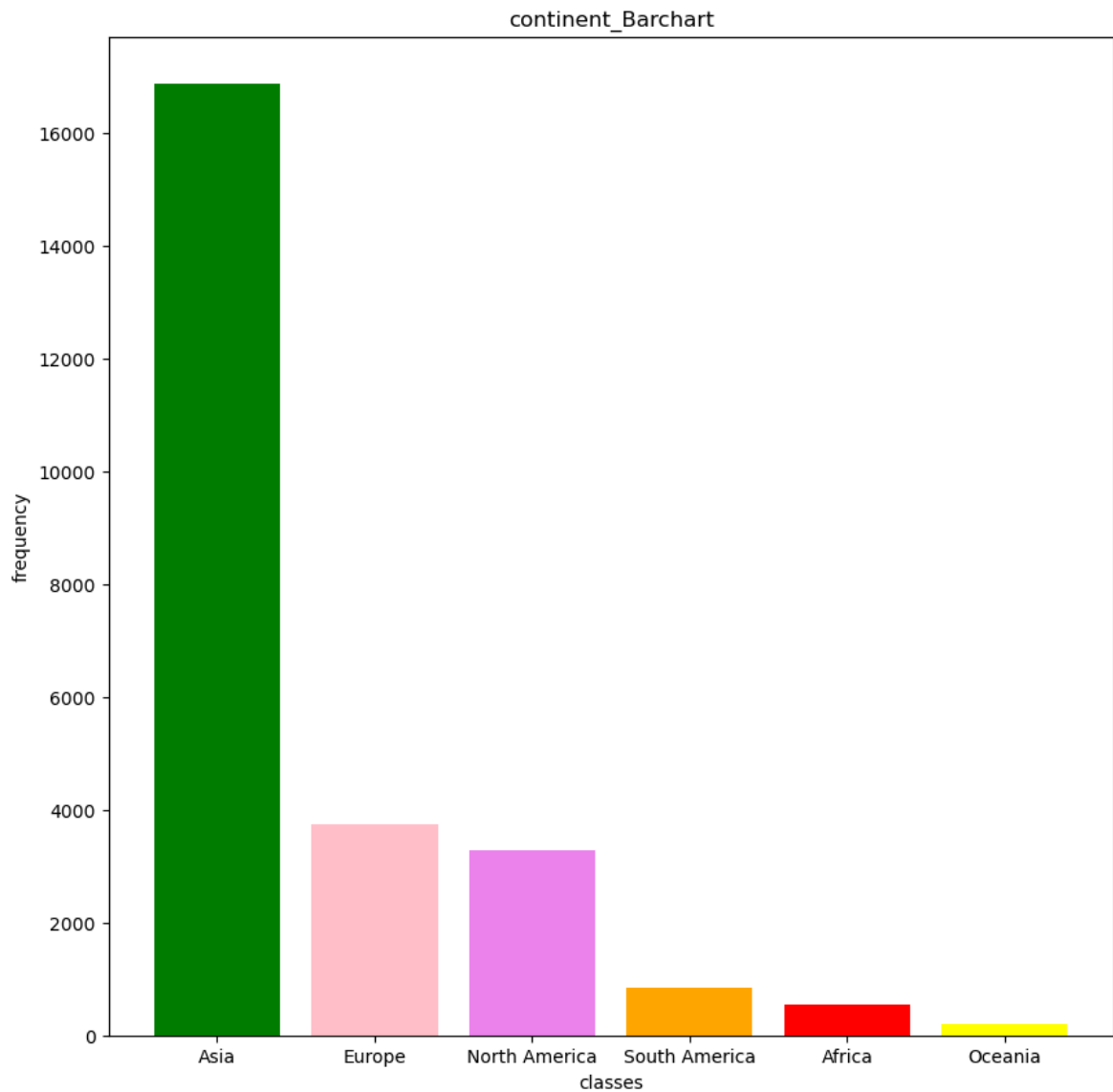
Out[40]: <Axes: >



```
In [41]: labels=visa_df['continent'].value_counts().keys()  
count=visa_df['continent'].value_counts().values
```

```
In [42]: # Bar chart code:  
  
plt.figure(figsize=(10,10))  
  
col=['Green','pink','violet','orange','red','yellow']  
plt.bar(labels,count,color=col)  
plt.xlabel('classes')  
plt.ylabel('frequency')  
plt.title('continent_Barchart')
```

```
Out[42]: Text(0.5, 1.0, 'continent_Barchart')
```



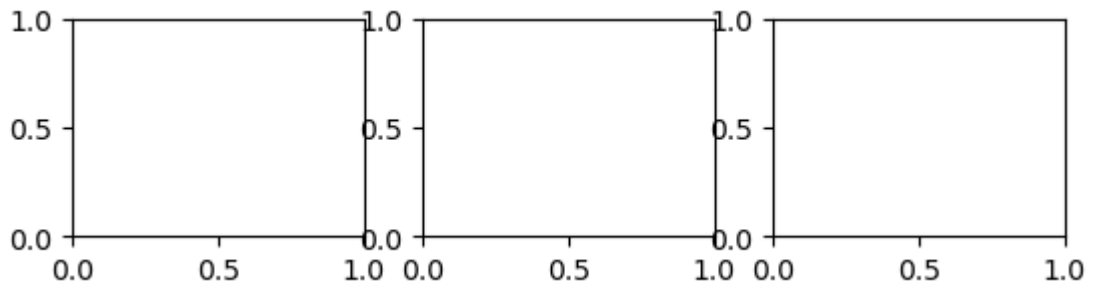
```
In [43]: # no-of categorical columns:9  
visa_df['case_id'].nunique()
```

Out[43]: 25480

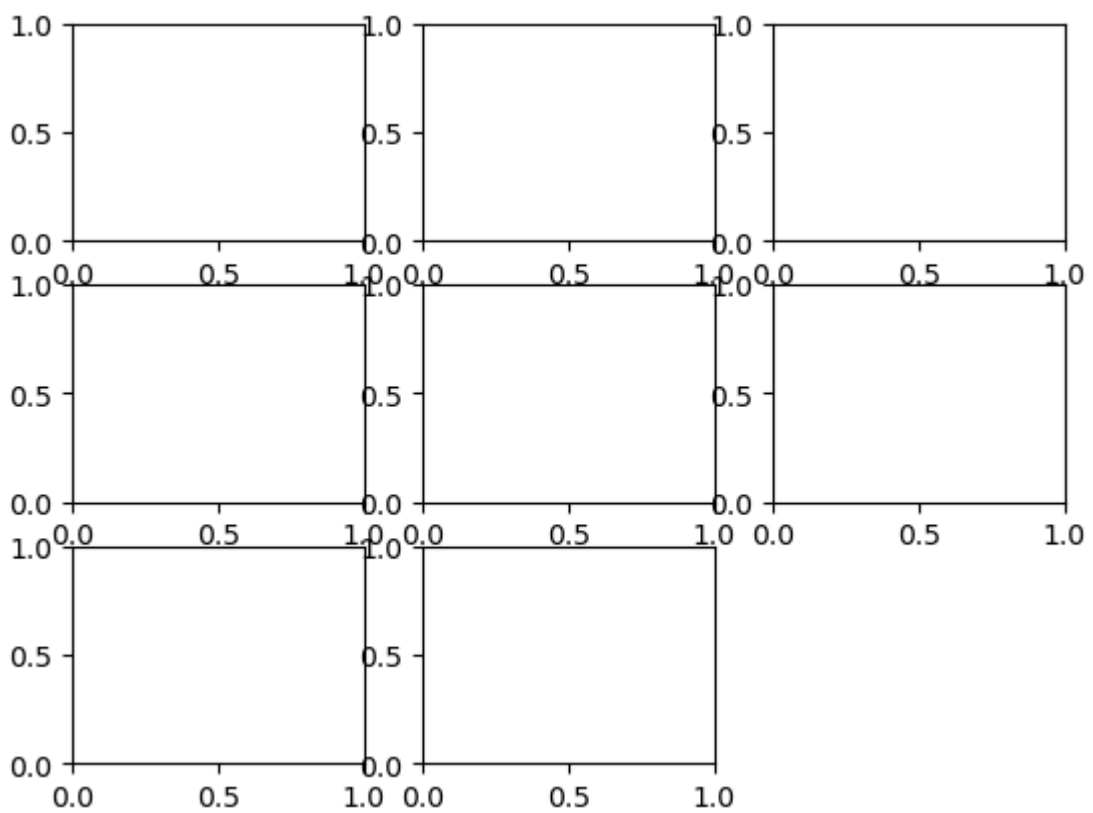
- no.of categorical columns:9
- in that we need to avoid case_id
- bcs case_id has 25480 different ids available
- there no -of categorical columns=8
- the possible matrix layout shape:
- 8x1 1x8 2x4 4x2 3x3

```
In [44]: # choosing 3x3 8 plots  
plt.subplot(3,3,1)  
plt.subplot(3,3,2)  
plt.subplot(3,3,3)
```

Out[44]: <Axes: >



```
In [45]: plt.subplot(3,3,1)
plt.subplot(3,3,2)
plt.subplot(3,3,3)
for i in range(1,len(cat)):
    plt.subplot(3,3,i)
```



In [46]: cat

```
Out[46]: Index(['case_id', 'continent', 'education_of_employee', 'has_job_experience',
               'requires_job_training', 'region_of_employment', 'unit_of_wage',
               'full_time_position', 'case_status'],
              dtype='object')
```

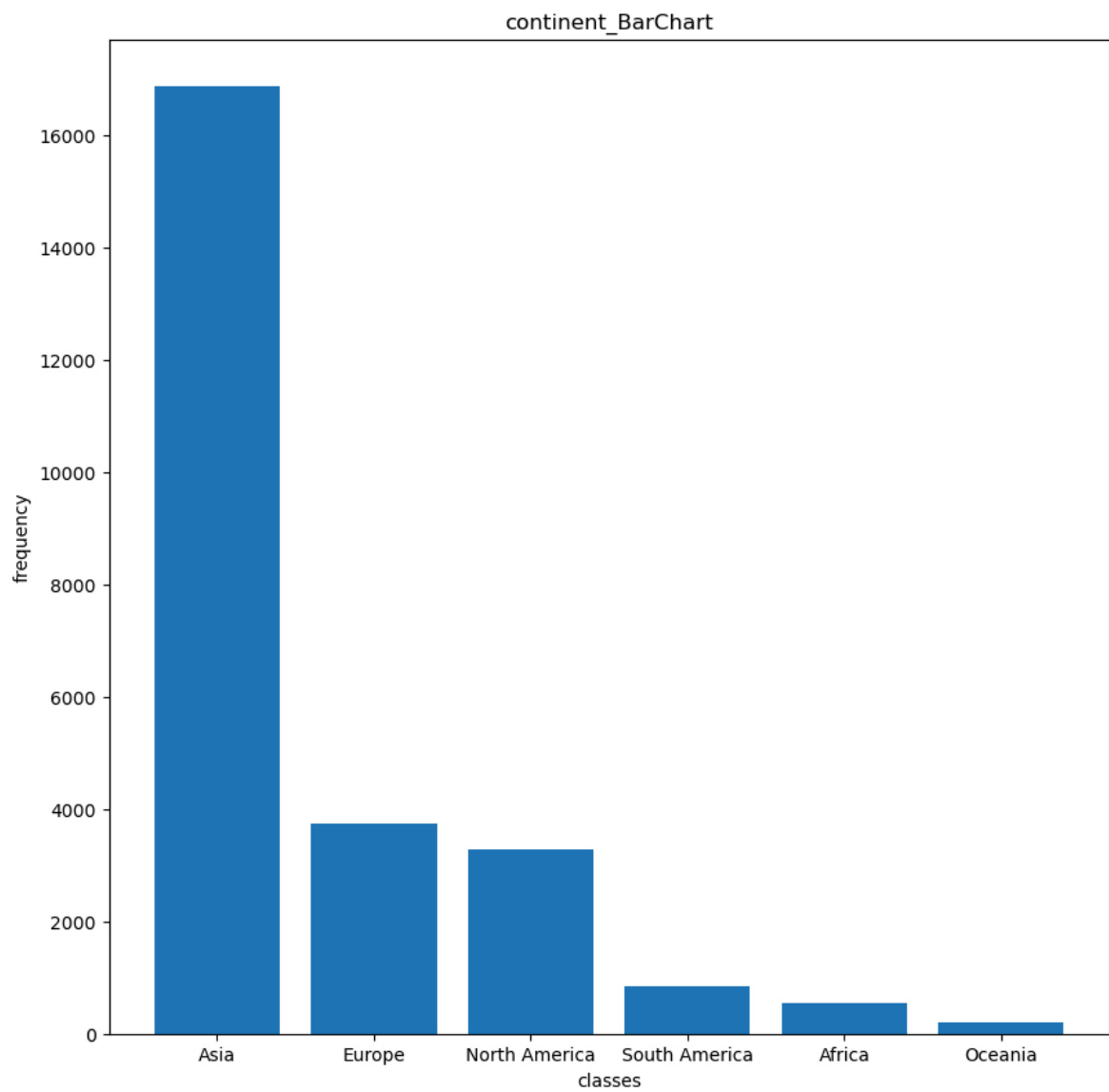
```
In [47]: for i in cat[1:]: # excluding case_id, starting from continent
```

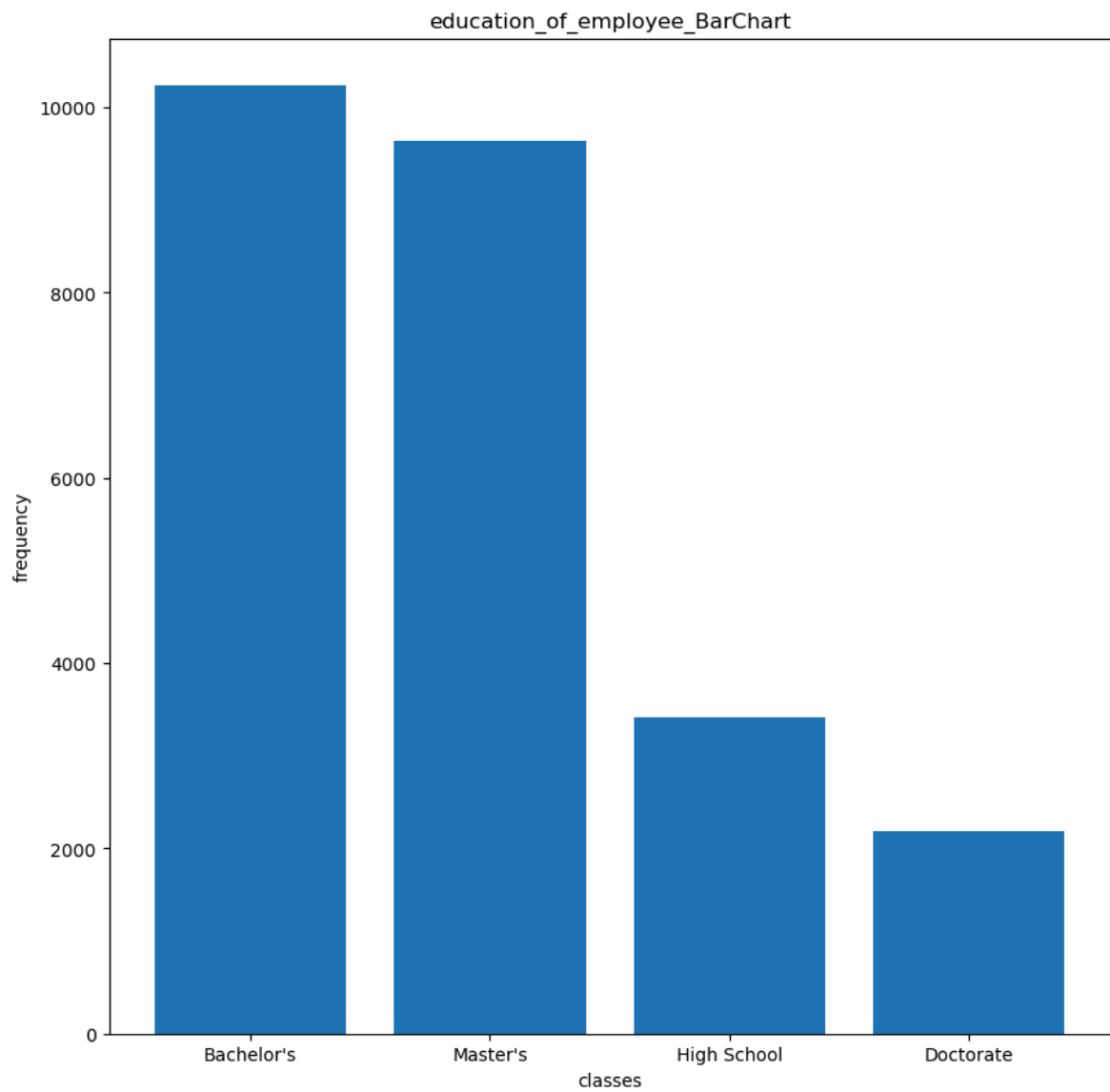
```
    keys=visa_df[i].value_counts().keys()
    values=visa_df[i].value_counts().values
```

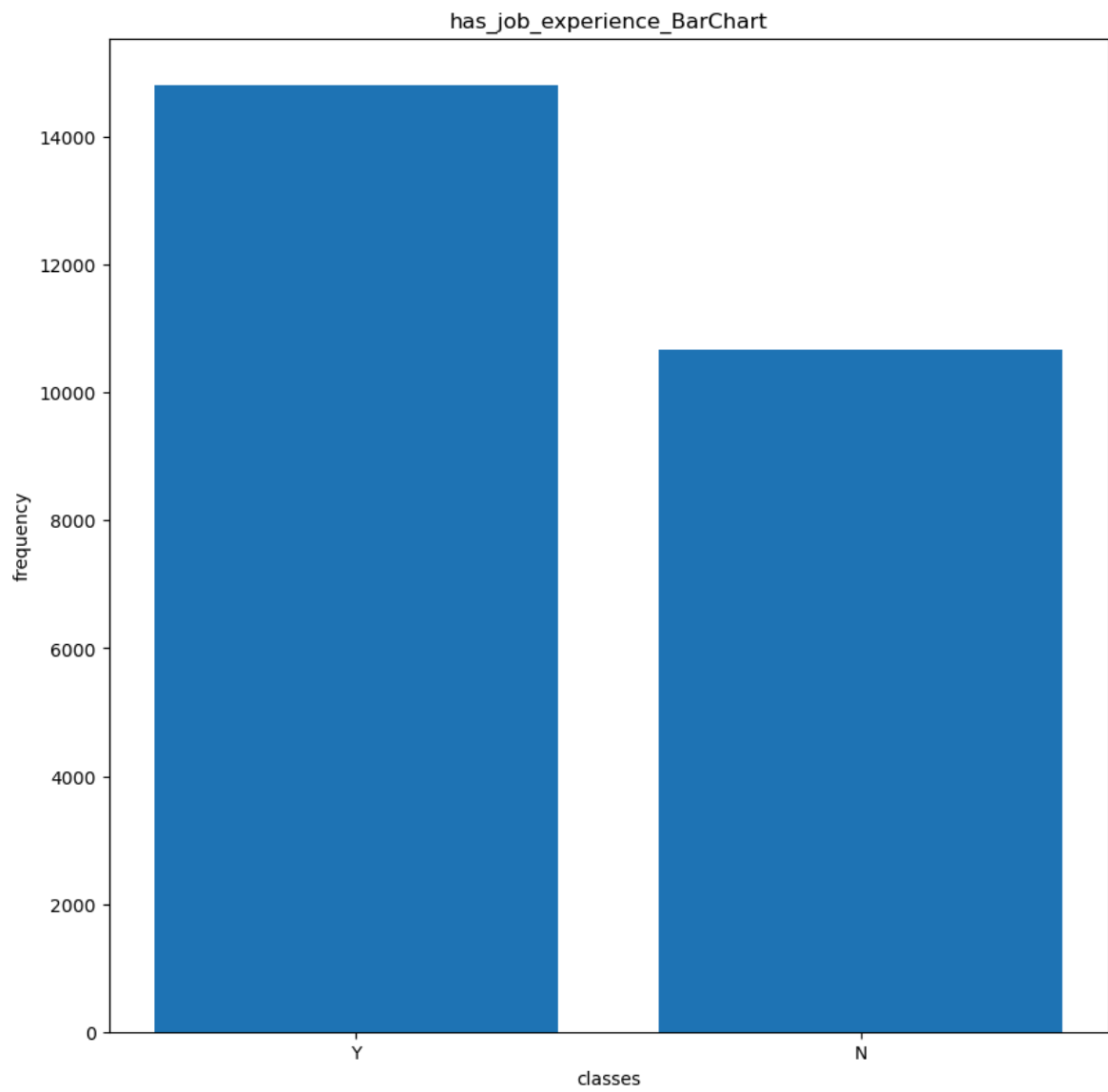
```
    plt.figure(figsize=(10,10))
```

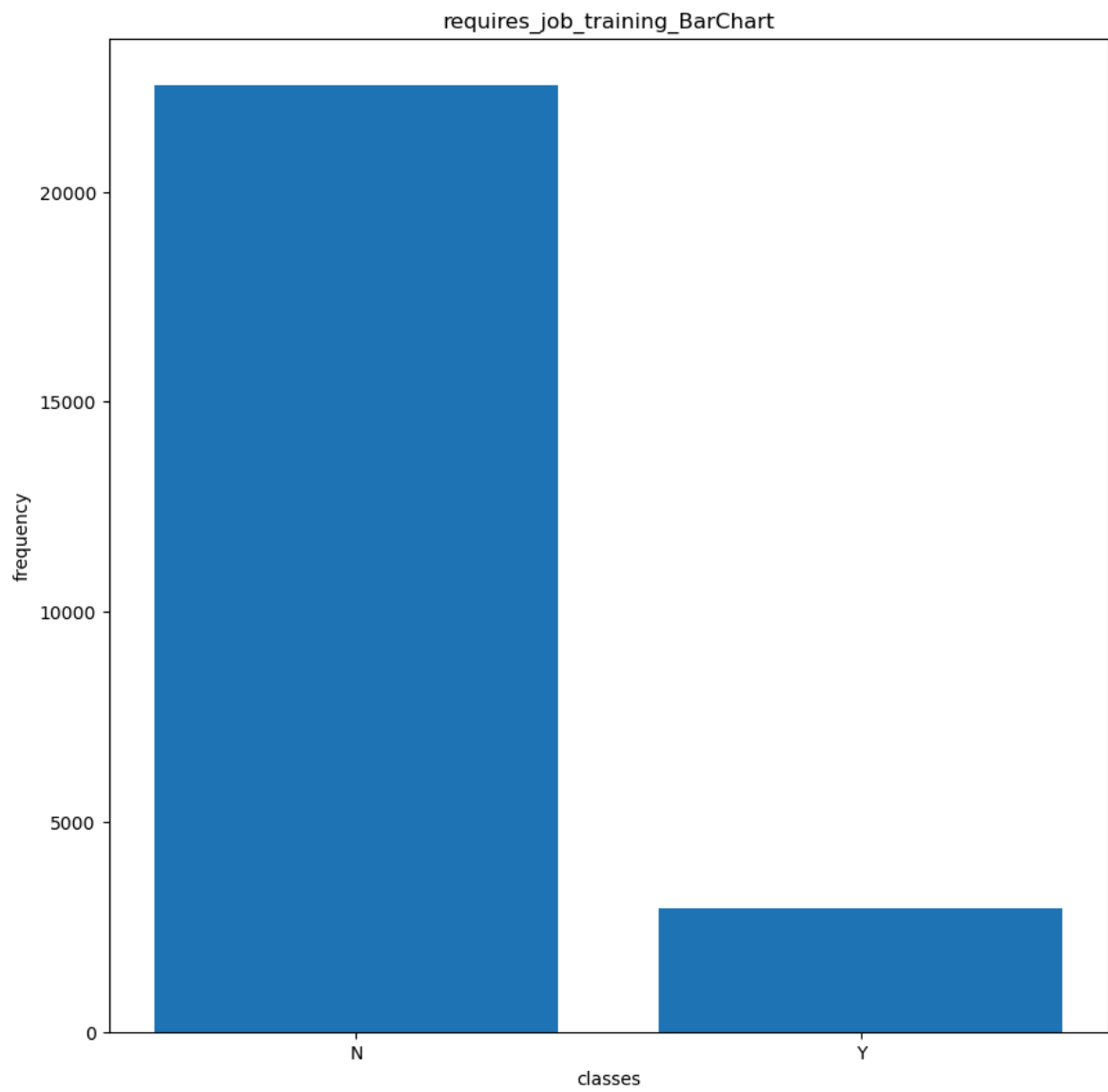
```
    plt.bar(keys,values)
    plt.xlabel('classes')
```

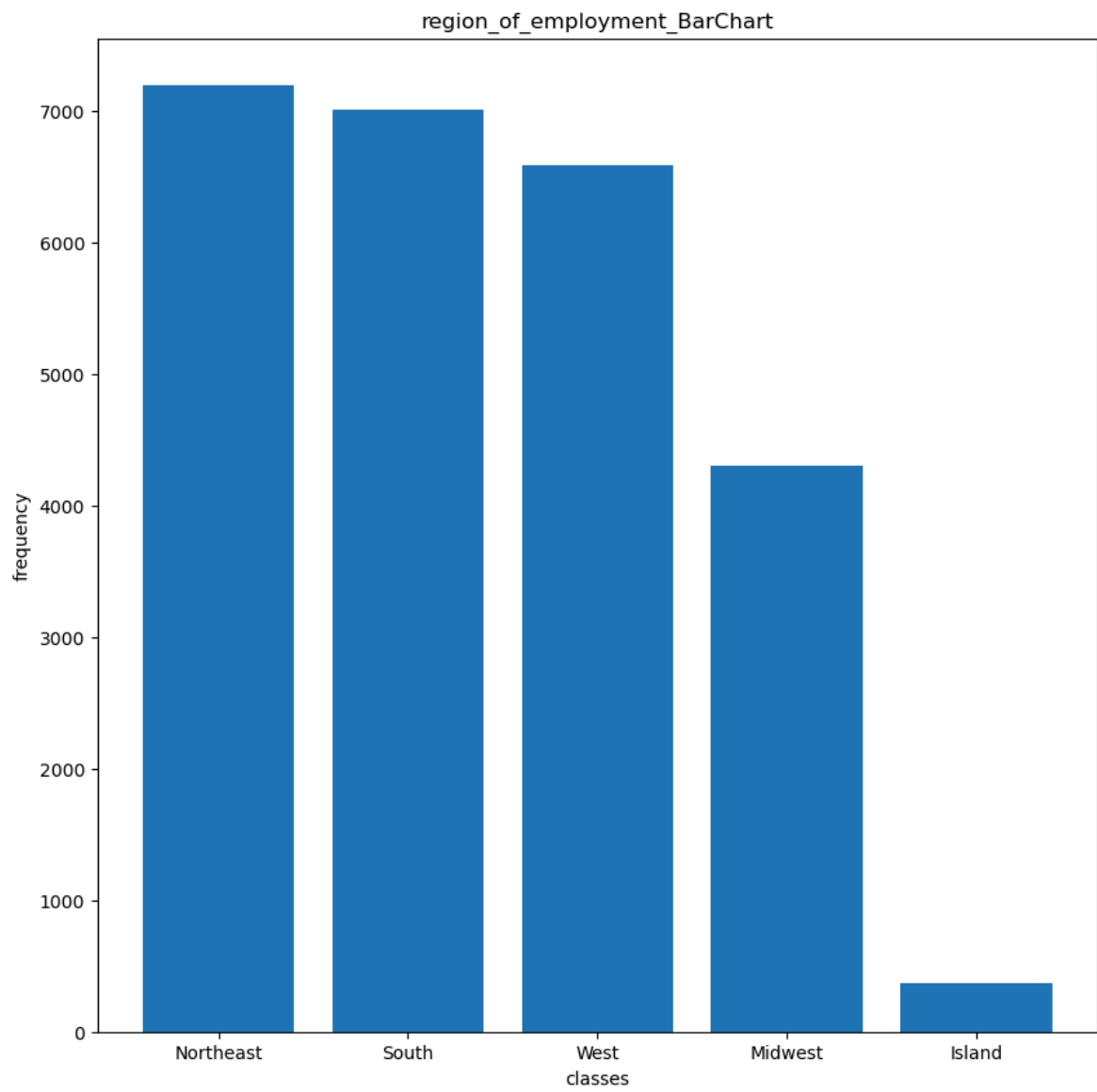
```
plt.ylabel('frequency')  
plt.title(f'{i}_BarChart')
```

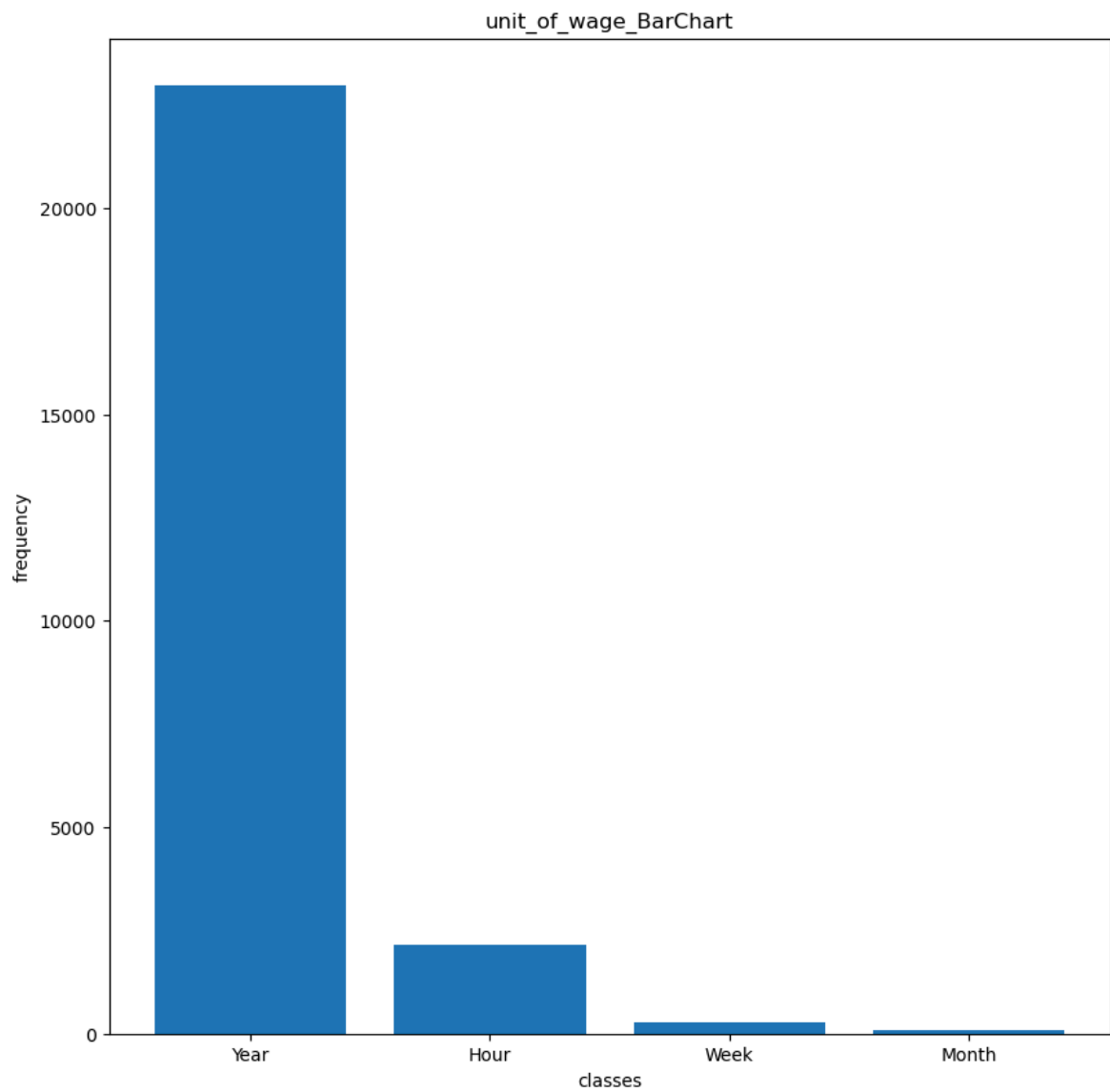


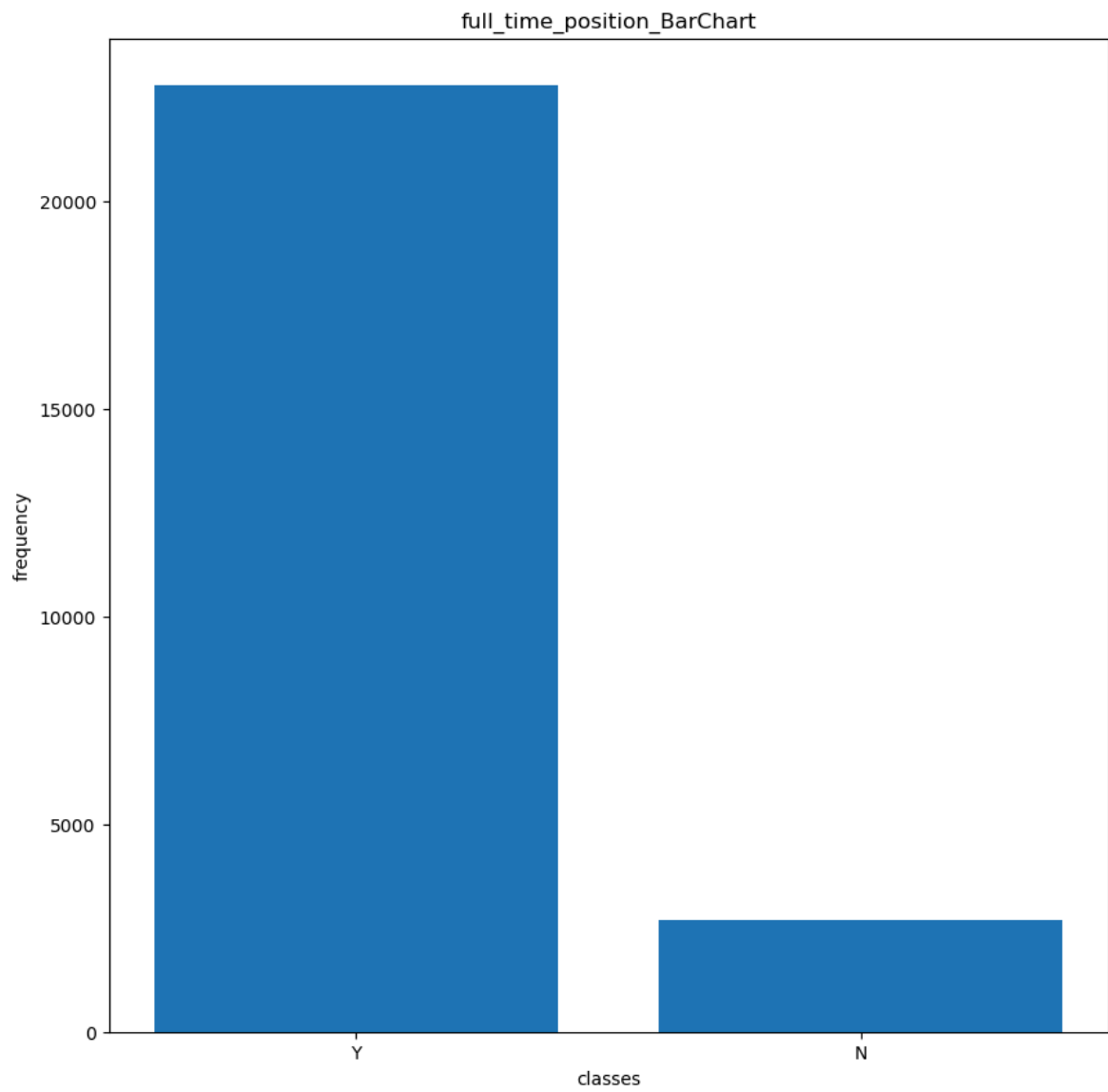


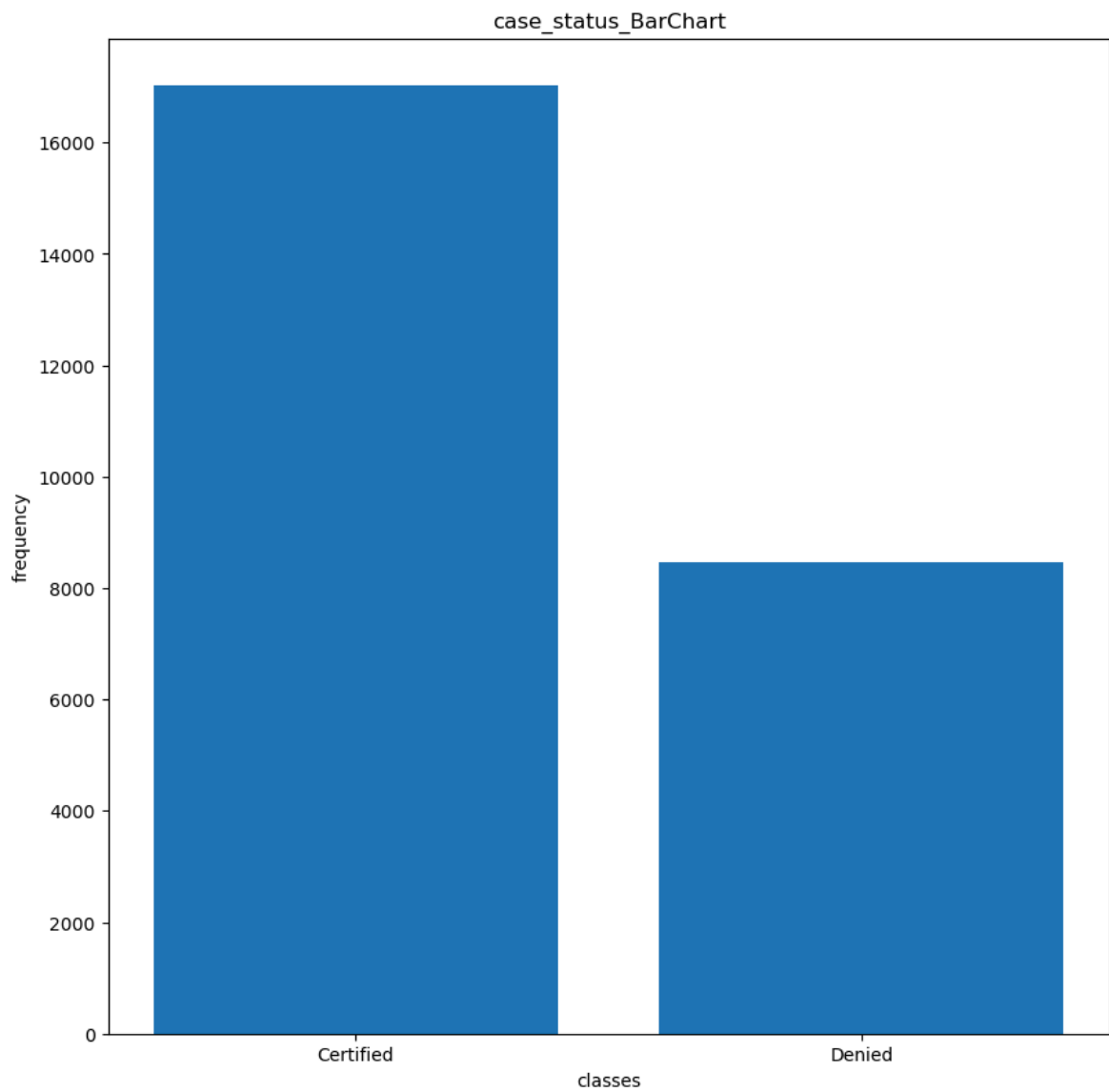






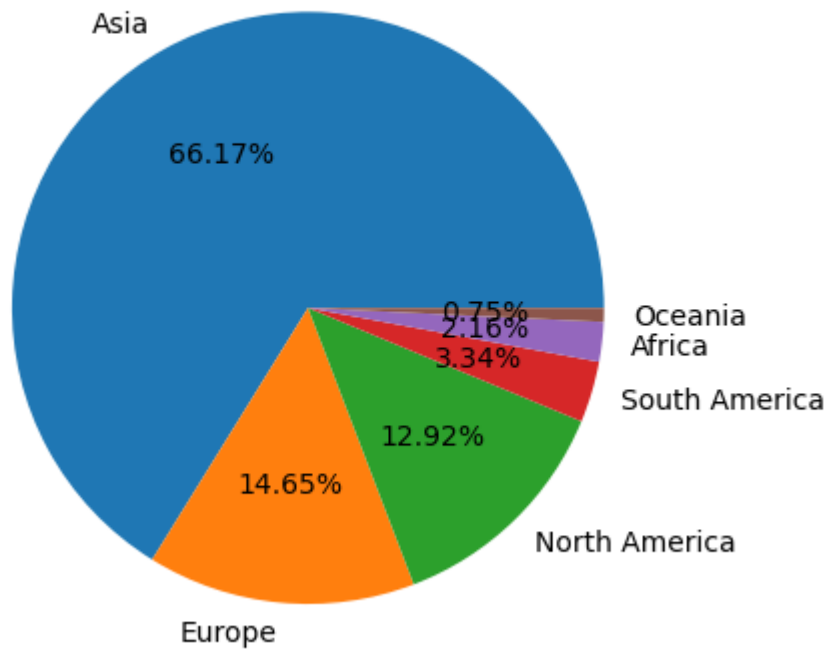




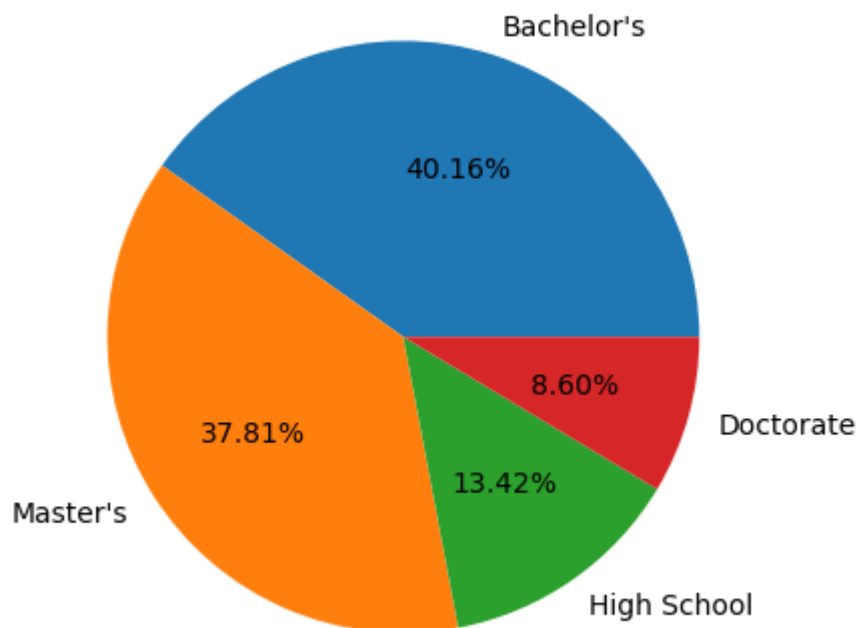


```
In [48]: for i in cat[1:]:  
         keys=visa_df[i].value_counts().keys()  
         values=visa_df[i].value_counts().values  
         plt.pie(values,labels=keys,autopct='%1.2f%%')  
         plt.title(f'{i}_piechart')  
         plt.show()
```

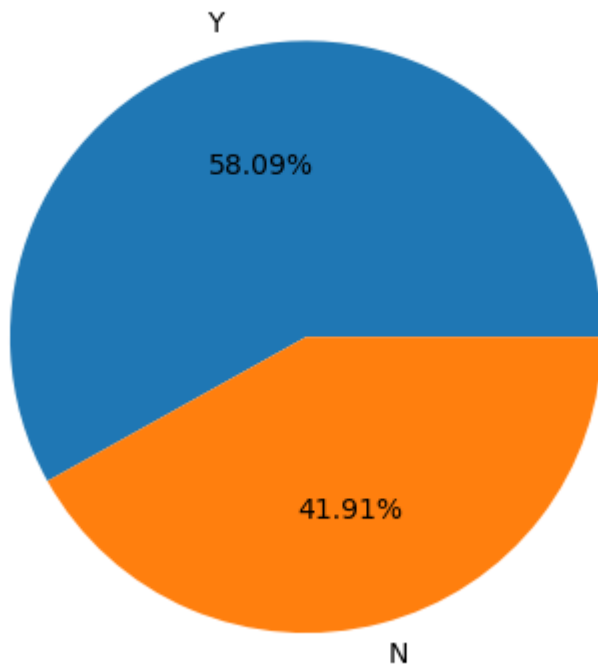
continent_piechart



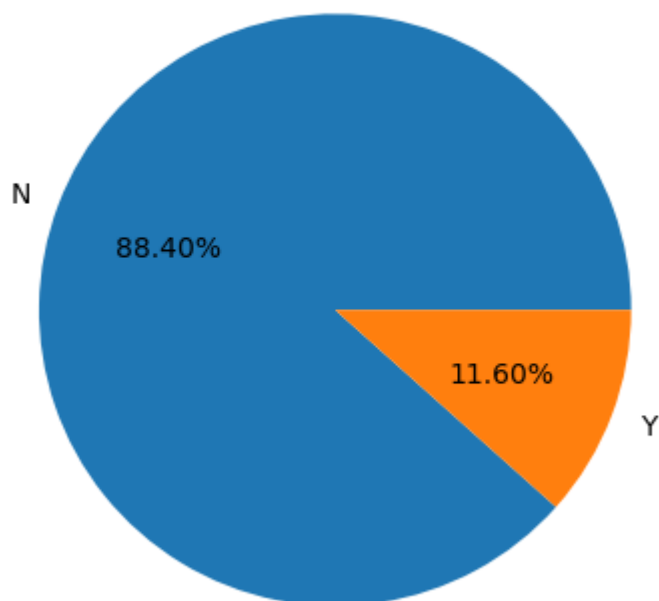
education_of_employee_piechart



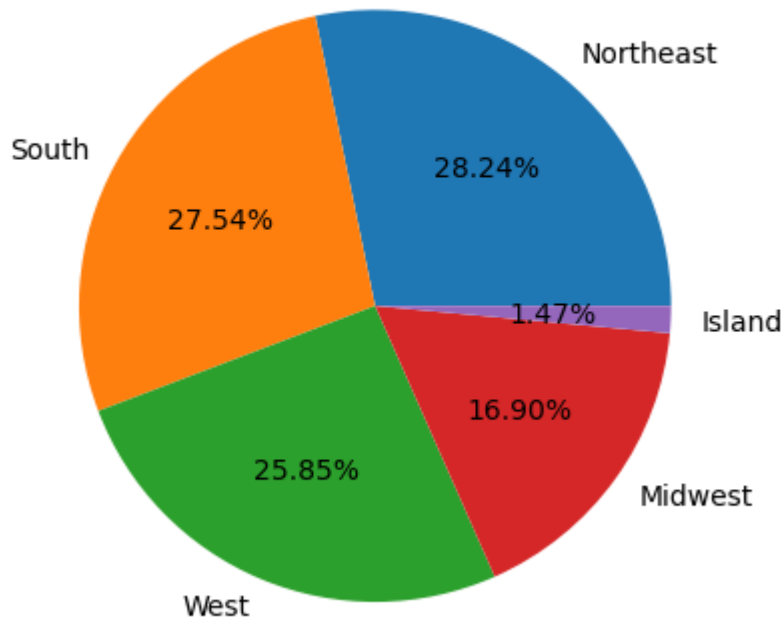
has_job_experience_piechart



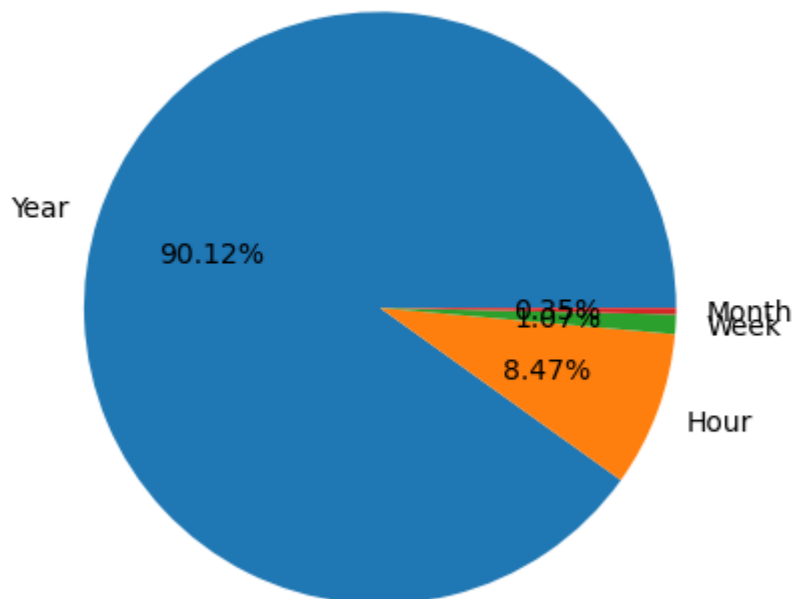
requires_job_training_piechart



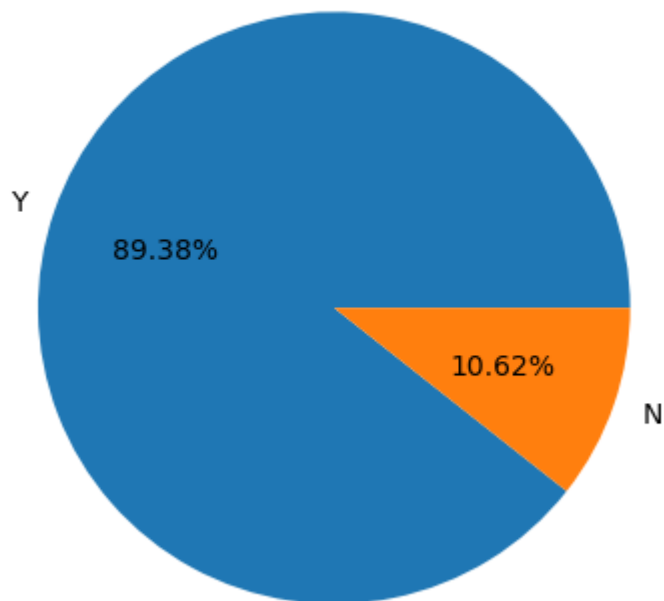
region_of_employment_piechart



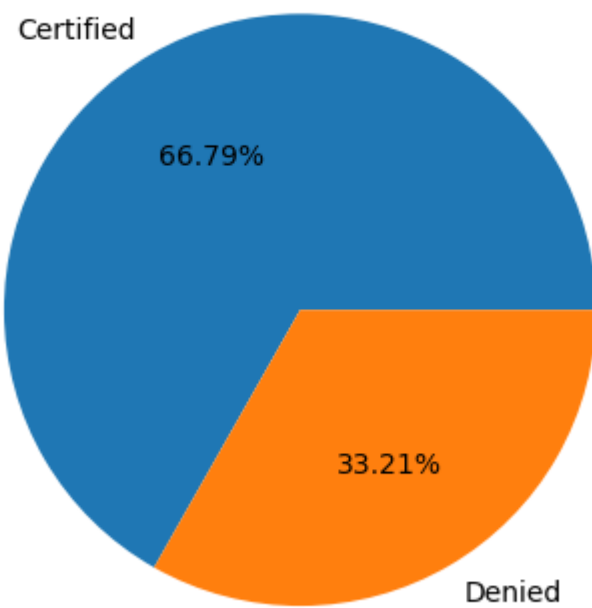
unit_of_wage_piechart



full_time_position_piechart



case_status_piechart



```
In [49]: for i,j in enumerate(cat[1:]):  
         print(i,j)
```

```

0 continent
1 education_of_employee
2 has_job_experience
3 requires_job_training
4 region_of_employment
5 unit_of_wage
6 full_time_position
7 case_status

```

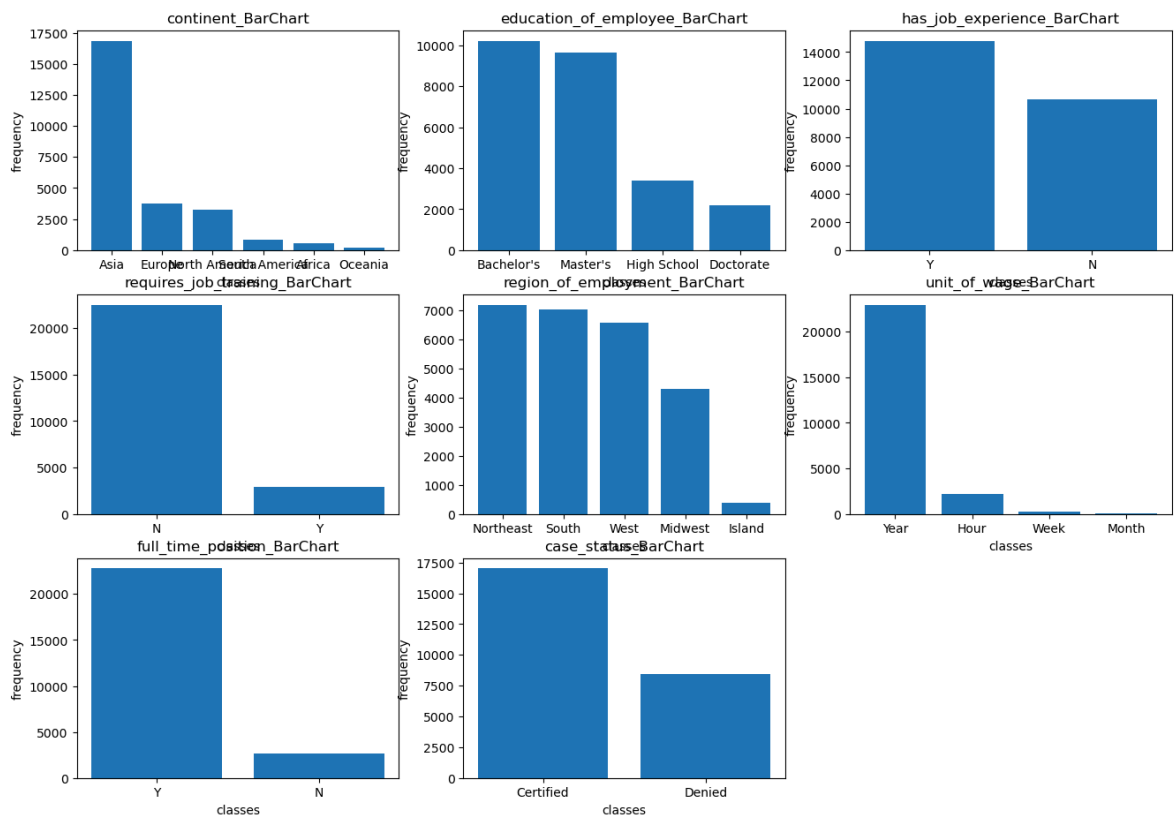
```

In [50]: #i ===== index , j====col name
plt.figure(figsize=(16,11))
for i,j in enumerate(cat[1:]):
    keys=visa_df[j].value_counts().keys()
    values=visa_df[j].value_counts().values

    plt.subplot(3,3,i+1)

    plt.bar(keys,values)
    plt.xlabel('classes')
    plt.ylabel('frequency')
    plt.title(f'{j}_BarChart')

```



Idea:

I want to save all the frequency tables

All the barcharts

All the pie charts

In separate folder automatically

Create all the frequency tables

```
In [51]: visa_df['continent'].value_counts().keys()
visa_df['continent'].value_counts().values
df=pd.DataFrame(zip(keys,values),
                 columns=['class','frequency'])
df.to_csv('continent_df.csv',index=False)
```

Step-1:Create a folder

```
In [52]: import os                                     # using try and except to avoid error when ru
folder_name='Frequency_table'
try:
    os.makedirs(folder_name)      # bcz folder already craeted
except Exception as e:
    print(e)
```

[WinError 183] Cannot create a file when that file already exists: 'Frequency_table'

```
In [53]: os.makedirs("Frequency_table",exist_ok=True) ## exist ok if created folder and r
```

Step-2:Save my df in frequency table folder

```
In [54]: visa_df['continent'].value_counts().keys()
visa_df['continent'].value_counts().values
df=pd.DataFrame(zip(keys,values),columns=['class','frequency'])
```

```
In [55]: # frequency_table path
#path=r
#df.to_csv('continent_df.csv',index=False)
```

```
In [56]: # directory=os.getcwd()
#folder_name='Frequency_table'
```

```
In [57]: # step-1 create a folder
folder_name='Frequency_table'
os.makedirs(folder_name,exist_ok=True)

#step-2: make a new path
directory=os.getcwd()
new_dir=os.path.join(directory,folder_name)

#step-3:MAKE a new location
```

```
file_name='continent_df.csv'
path=os.path.join(new_dir,file_name)
path
```

Out[57]: 'C:\\Users\\Lenovo\\Frequency_table\\continent_df.csv'

In [58]: df.to_csv(path,index=False)

In [59]:

```
#step-1:
folder_name='Frequency_table'
os.makedirs(folder_name,exist_ok=True)

#step-2:make a path=====

directory=os.getcwd()
new_dir=os.path.join(directory,folder_name)

#step-3:create a Data frame=====
for i in cat[1:]:
    visa_df[i].value_counts().keys()    #i=columns,before 'i'===continent
    visa_df[i].value_counts().values
    df=pd.DataFrame(zip(keys,values),columns=['class','frequency'])

#step-4:Make a new path=====
file_name=f'{i}_df.csv'
path=os.path.join(new_dir,file_name)

#step-5:save the DataFrame
df.to_csv(path,index=False)
```

Same for BAR chart And pie chart

In [60]:

```
#Step-1:
folder_name='Frequency_table'
os.makedirs(folder_name,exist_ok=True)

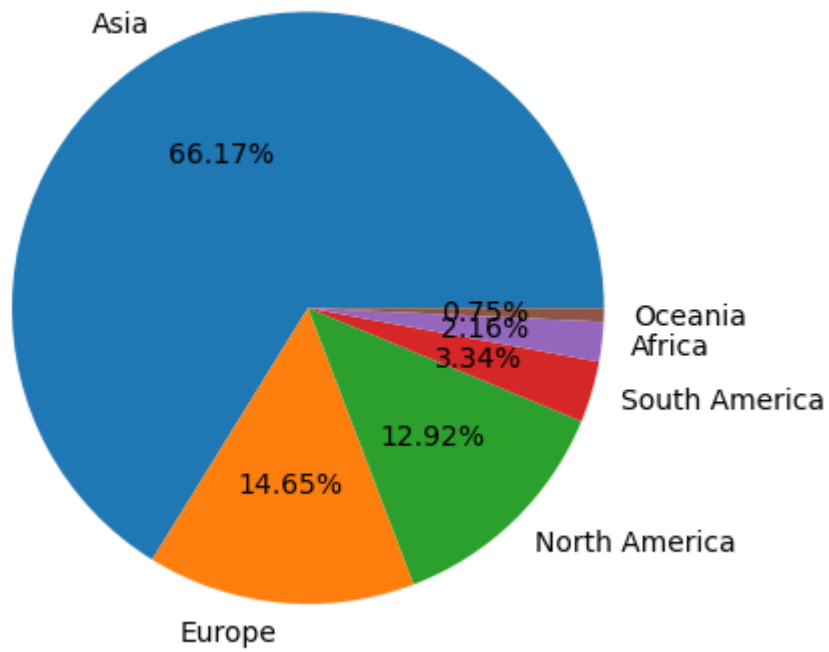
#Step-2:
directory=os.getcwd()
new_dir=os.path.join(directory,folder_name)

#Step-3
for i in cat[1:]:
    keys=visa_df[i].value_counts().keys()
    values=visa_df[i].value_counts().values
    plt.pie(values,labels=keys,autopct='%1.2f%%')
    plt.title(f'{i}_piechart')
    plt.show()

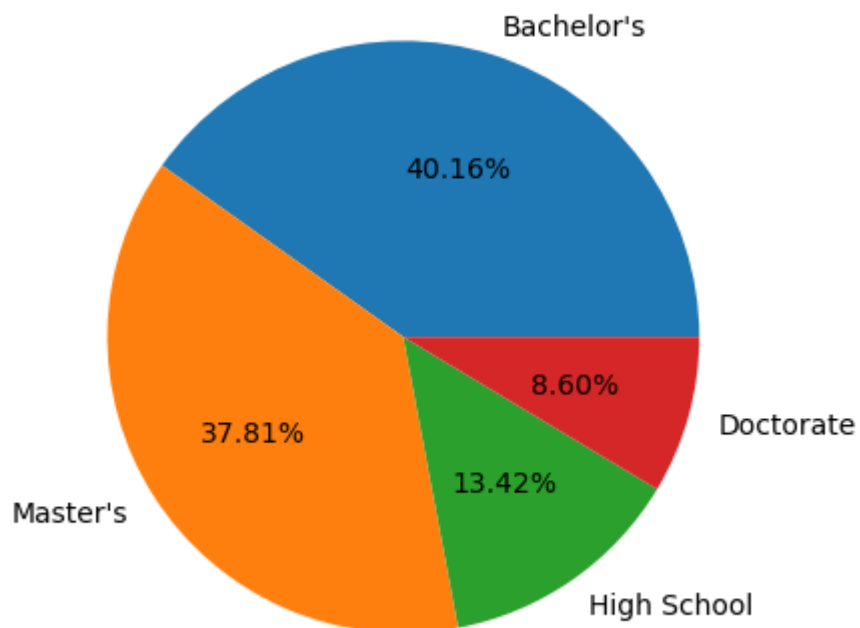
#Step-4
file_name=f'{i}_df.csv'
path=os.path.join(new_dir,file_name)

#Step-5:
df.to_csv(path,index=False)
```

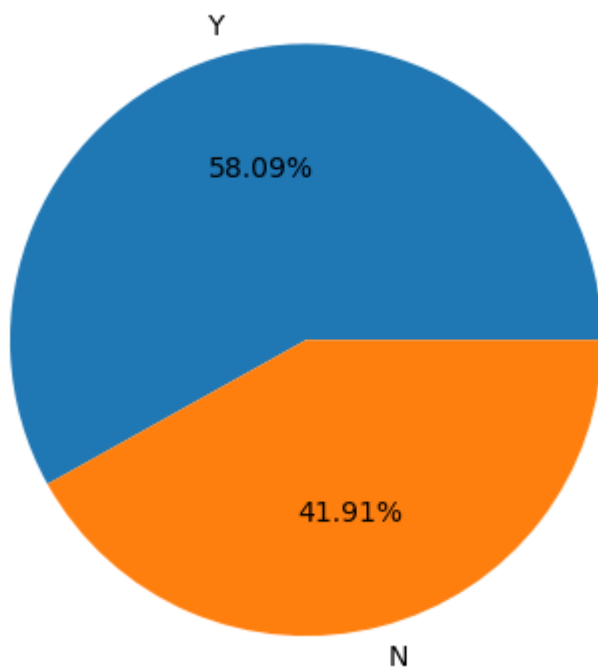
continent_piechart



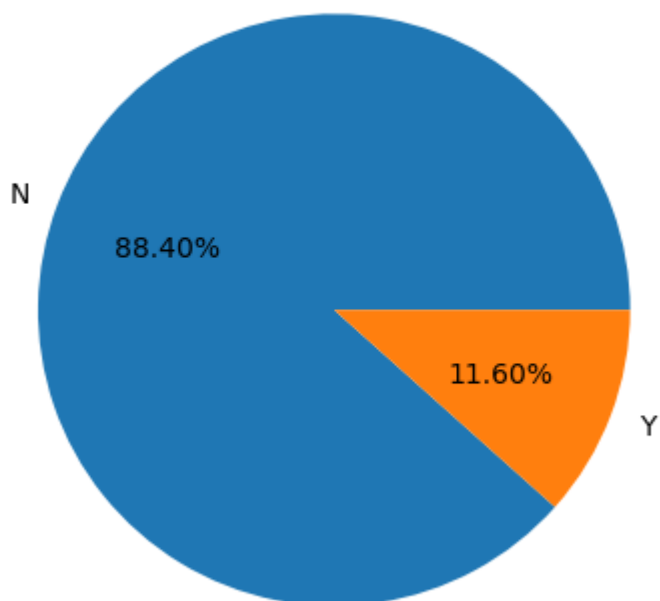
education_of_employee_piechart



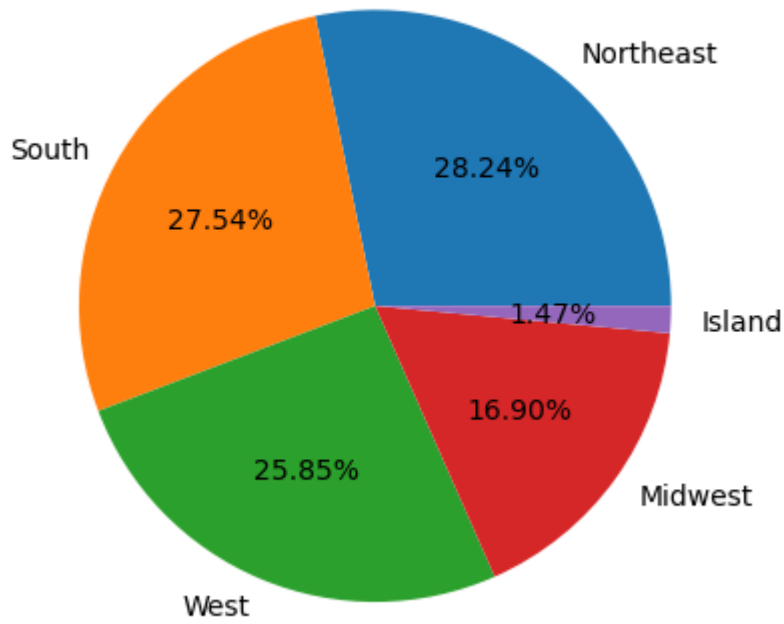
has_job_experience_piechart



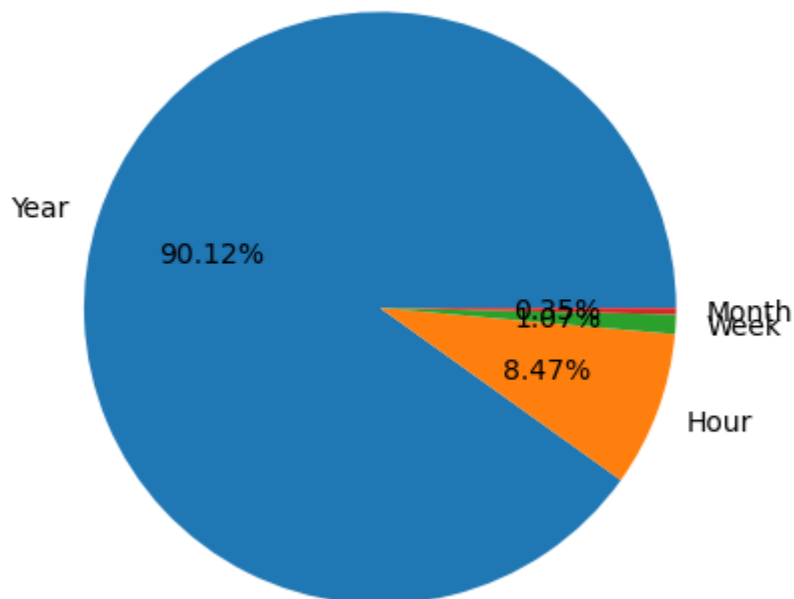
requires_job_training_piechart



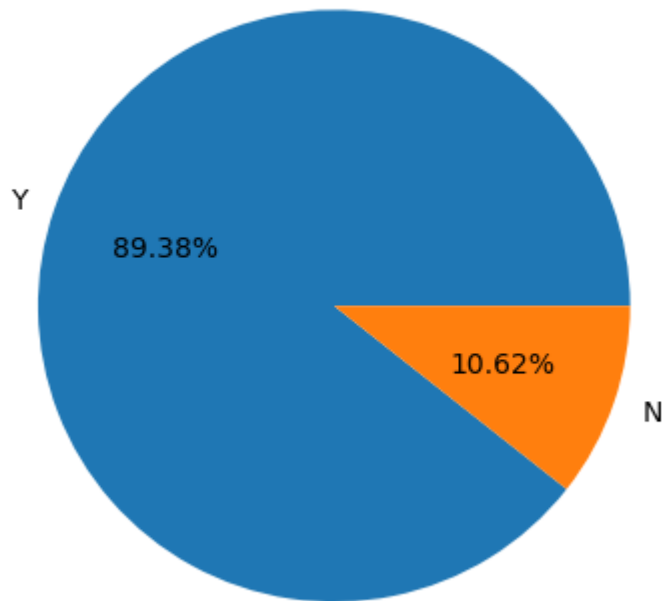
region_of_employment_piechart



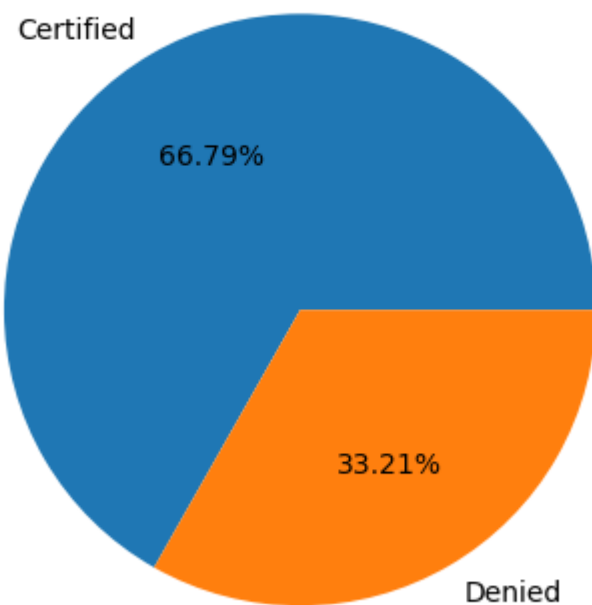
unit_of_wage_piechart



full_time_position_piechart



case_status_piechart



```
In [61]: #step-1:
folder_name='Frequency_table'
os.makedirs(folder_name,exist_ok=True)

#step-2:
directory=os.getcwd()
new_dir=os.path.join(directory,folder_name)

#step-3
for i in cat[1:]: # ecluding case_id,starting from continent
```



```
keys=visa_df[i].value_counts().keys()
values=visa_df[i].value_counts().values

plt.figure(figsize=(10,10))

plt.bar(keys,values)
plt.xlabel('classes')
plt.ylabel('frequency')
plt.title(f'{i}_BarChart')

#step-4
file_name=f'{i}_df.csv'
path=os.path.join(new_dir,file_name)

#step-5:
df.to_csv(path,index=False)
```

